The Switch: Determinants of family school modality choices during the COVID-19 pandemic

An Honors Thesis submitted in partial fulfillment of the requirements for Honors Studies in
Fulbright College of Arts and Sciences

By
Steicy Lopez
Economics
Spring 2025
The University of Arkansas

Abstract

The K-12 system in the United States experienced substantial challenges during the COVID-19 pandemic. This study aims to give an overview of public, private, and homeschooling enrollment decisions throughout this period, the changes families made between types of schooling, and identify familial characteristics associated with the probability of switching in and out of public education. Knowing more about the groups exiting and returning to the public school system can help public school districts better tailor policies to attract and retain their students. Socioeconomic factors influencing a change could be signs of increasing segregation. Using discrete choice duration models and multinomial logistic models, we find that education, income, and trust in Fox News were among the most important characteristics related to predicting school type decisions throughout the pandemic. We then document the changes in the importance of these factors over time.

Introduction

Between 2020 and 2022, public K-12 schools lost more than 1.2 million students (Dee, 2023), according to administrative records. Accounting for approximately 3% of students in the US (Zamarro & Camp, 2025), this loss contrasts with a pre-pandemic annual growth rate of enrollment in public schools of 0.5 percent (National Center for Education Statistics, 2023).

The loss in public school enrollment is concerning given enrollment numbers are often the biggest source of revenue for public schools. This loss in revenue could lead to public school closings, the cutting of programs, or personnel layoffs. Considering that layoffs are often based on revenue cuts rather than performance, these outcomes can deteriorate the quality of public schooling (Gordon, et al., 2018). Han et al. (2017) found that while causing psychological harm to families, schools that closed between 2006-7 and 2012-13 typically experienced a loss in enrollment before having to close. Public schools often offer resources to the community beyond schooling, like free and reduced lunch, extracurricular activities, and access to technology that students might be losing access to if public schools experience budget cuts or closings.

The Census Bureau Pulse Survey (2021) found that the percentage of Black families homeschooling increased by more than 5 times from April 2020 to December 2020. If the probability of families making the switch is associated with racial, ethnic, or financial characteristics, the system could see an increase in socioeconomic segregation. Coupled with the deterioration in public schools due to revenue cuts, such segregation could disrupt educational progress for minority groups. According to a US Government

Accountability Office (GAO) report, more than a third of students already attend schools where 75% or more of students identified in a single race category despite the growing diversity of students nationally (2022). This invites further questions about whether these students have returned now and how these characteristics played into their decisions.

Additionally, the time that students spent away from public schools could point to differentiation in educational experiences as some evidence points to lower outcomes for those in public school compared to private options during the pandemic (ERB, 2024). A CATO Institute survey found that while 88 percent of private schools reopened to in-person instruction by Fall 2020, it was not until April 2021 that about half of public schools had offered the same (McCluskey & Ekins, 2024). Tracking the types of families that were more likely to leave could be useful in understanding disparities in educational outcomes even if they did return to their pre-pandemic school type.

After accounting for private and home-schooling growth and population loss, more than 1/3 of the public school enrollment decline remains unexplained (Dee, 2023). Dee (2023) theorizes this could be due to an increase in truancy, higher rates of unregistered homeschooling, and an increase in skipping kindergarten. Any of these trends are likely to have negative effects on the educational system at large as well as outcomes of individual students. Thus, understanding this gap can motivate different policy responses. Population loss due to demographic shifts appears persistent, though there are signs of reversal. Many states experienced declines as people relocated or changed jobs during the pandemic (United States Census Bureau, 2023). These shifts may have inflated public school enrollment loss rates. However, since the initial shock of the pandemic, many families appear to be returning, making the persistence of enrollment gaps puzzling.

Lastly, the school choice movement in the United States has strengthened in recent years with many states passing legislation supporting the marketization of schools (Abdulkadiroglu & Andersson, 2022). Despite these changes, little is known about how parents and guardians respond to increased options and the growth of competitive pressures in education. Examining parental decision-making during crises, such as the pandemic, and the persistence of these choices can shed light on potential risks when implementing various education policies.

Literature Review

Various factors have been tied to higher declines in public school enrollment such as grade level, urbanicity, income, and ethnic/racial background. Additionally, many of these factors have been associated with school modality decisions during the pandemic like in-person, virtual, or hybrid schooling. To further complicate matters, Bacher-Hicks et al. (2014) found that the stickiness of these decisions does vary by school type pursued. In Michigan, only 21 percent of students who switched to private schools in Fall 2020 returned by Fall 2021, compared to 50 percent of those who switched to homeschooling. This suggests there are substantial differences between private and homeschooling switchers. This study attempts to account for these and additional factors that could be confounding variables when analyzing these characteristics while using a longer run, nationally representative sample.

The share of K-12 students attending elementary schools dropped 6.8 percent from 2019-20 to 2020-21 and witnessed a small recovery in 2021-22 (-6.3 percent relative to 2019-20) nationally. Interestingly, middle schools saw a 3.3 percent and a 5.3 percent drop respectively (2019-20 to 2020-21 and 2020-21 to 2021-22), while high schools experienced limited changes in enrollment levels throughout the pandemic (Burtis & Goulas, 2023). The dynamics behind these declines are emphasized as elementary and middle school ages are crucial in educational development.

Urbanicity also impacted the public enrollment declines across the country. Urban students who exited public schools are showing a greater resistance to returning to public school while rural districts recovered significantly (Bacher-Hicks, Musaddiq, Goodman, & Stange, 2024). Burtis and Goulas (2023) found that elementary school enrollment relative to 2019-2020 as of 2021-2022 decreased by approximately 9% in urban areas while enrollment in rural areas experienced only a 3% loss nationally. Urbanicity has been tied to schooling decisions as the availability of schooling options in urban areas and access to the internet could influence a family's interest in untraditional education options. Further, Camp and Zamarro (2021) and Camp, Johnson, and Zamarro (2023) find that living in a rural area had a significant positive effect on attending school in person in Fall 2021.

Socioeconomic sorting during the pandemic through school choice could pose risks to equity in education. Evidence of this is already being tracked by Dee and Murphy (2023) where it was found that public-school enrollment declines have been most severe in smaller traditional districts, districts with higher concentrations of poverty, and districts with higher concentrations of white students. Income has historically been associated with various schooling decisions. Murname & Reardom (2018) found that

private school enrollment has been dropping among middle-class families for the last five decades while high-income family enrollment in private schools has stayed consistent.

During the COVID-19 pandemic, families experienced sharp and perhaps unexpected changes in income. Meanwhile, stimulus checks could have provided some economic stability and potentially mobilized families to pursue paid schooling options. Importantly, Bacher-Hicks et al. (2024) finds that public schools can expect to continue seeing declines in enrollment especially from higher income groups. Another tie from income to schooling decisions during the pandemic is established by Calarco, Coleman, & Halpern-Manners (2021) where they show lower income families were found to be more likely to have pursued virtual learning during the pandemic.

Similarly, the COVID-19 pandemic had disproportionate effects on different racial or ethnic groups that could have led these families to seek alternative types of schooling. During the first ten months of the pandemic, there was a significantly higher proportion of Hispanic/Latino hospitalizations relative to the general population in all US census regions. Black citizens in the southern region experienced the 2nd highest proportion of hospitalizations. Although Black hospitalization rates saw upticks by December 2020, ethnic and racial disparities overall seemed to have gradually decreased throughout the pandemic (Romano, et al., 2021).

Decisions on COVID-19 policy and critical race theory in public schools were highly politicized during the pandemic (Shao & Hao, 2020). Reopening policy was found to be more associated with political leanings than COVID-19 severity. More than a third of teachers and most principals reported getting harassed "because of their school's policies on COVID-19 safety measures or for teaching about race, racism, or bias" in the 2021-22 school year (Woo, et al., 2022). By the end of the 2021-22 legislative session, 16 states had passed critical race theory bans (Kelly, 2023). As these bans continue to be contested across the US, parental views on this issue could influence a switch out of public school. Miles (2021) cites reasons such as the use of homeschooling as a tool for activism, avoiding educational racism, and incorporating racial history in their children’s learning to explain the increase in the proportion of Black families homeschooling. Investigating how these leanings materialized into school choice decisions during the pandemic will be productive for school choice literature and could point to significant polarization risks.

Considering the shift in enrollment between virtual, hybrid, and in-person learning during the pandemic, parents may have moved students out of the public school system due to the availability of in-person,

remote, or hybrid learning at their current schools. Decisions after May 2021 would not be reflected by modality options as nearly all school districts offered either hybrid or in-person learning by this time (Institute of Education Statistics, 2021). Our analysis considers these factors and additionally, due to the complexity and potential intersectionality of these variables, we include factors in our analysis like working status, perceived COVID risk, number of kids in the household, media trust, institutional trust, and medical status in our set of explanatory variables.

Specifically, the central questions in this research are:
What changes in enrollment happened between public, private, and homeschooling during and after the COVID-19 pandemic?

What characteristics of families are most associated with switching in and out of public schools during this time?

Data and Descriptive Statistics

This study employs data from the Understanding America Study (UAS) conducted by the University of Southern California (USC). A branch of this study, the Understanding Coronavirus in America Study (UCA), consists of a nationally representative panel of households[1] that answered internet surveys throughout the pandemic. Respondents answered questions about the educational experiences of a randomly selected student living in their household. Cases where the same respondent reported inconsistent schooling levels (e.g., high school followed by elementary) were dropped, as these likely reflect changes in child selection due to graduating from school rather than actual transitions. Additionally, respondents with no children were removed from the sample. We make use of 14 UCA data sets and 5 additional post-pandemic waves (UAS 461, UAS 475, UAS 502, UAS 559, UAS584) comprising an average of 1,692 respondents per wave. The sample consists of 3,975 unique respondents and 33,832 total observations. Political leanings and urbanicity were also merged into the data set from the UAS U.S. Election Longitudinal study.

The analysis is divided into four phases reflecting changes in regulations, vaccine availability, and learning modality options available. A table of these phases as well as associated survey waves can be found in **Appendix A**. During Phase 1 (from April 2020 to October 2020), public schools began to reopen

[1] Weights were not used in regression analysis due to model complexity, but we believe our results would not differ substantially due to similar weight/non-weighted tabulations. See **Appendix D.**

and offer different attendance modes. However, these offerings were not consistent nationwide and varied widely from district to district. We account for this by constructing an exposure to remote learning variable that serves as a measure of remote learning supply. Further details surrounding this variable can be found in **Appendix B.** For alluvial diagrams, we divide Phase 1 into two phases cutting off Phase 0 in September 2020. This split off is solely used in these diagrams and the full stretch of Phase 1 between April 2020 and October 2020 is used in all analyses. In Phase 2, beginning at the end of December 2020 and ending January 2021 public schools were increasingly offering in-person learning options. Due to this, we expect that the availability of in-person learning became a diminishing factor in school choice decisions from this point onward. By Phase 3 (between September 2021 and May 2022), nearly all schools were offering in-person instruction. Lastly, Phase 4, spanning from July 2022 to February 2024, marks a period where attitudes surrounding COVID-19 are expected to play a less significant role in explaining school choice decisions.

School Type Variable

As the pandemic shocked educational systems, it was not clear how to best track students. Across survey waves, there was not a consistent way school type questions were asked. For most waves we employed, families were asked about their school type in two questions.

Is [selectedchild] enrolled in a public, private, charter or virtual school for the 2020-21 school year (even if currently attending school from home), or are they homeschooled, or does some other situation apply?

1 Yes, [] is enrolled in a public, charter, private or virtual school (select this option even if your child is attending school from home)

2 No, [] is not enrolled and is homeschooling

3 No, [] is neither enrolled nor homeschooling (e.g. dropped out, taking a year off, pursuing a GED)

4 No, some other situation applies (please specify):

What kind of school is [selectedchild] enrolled in?

1 Public school

2 Private school

3 Charter school

4 Virtual School

5 Other

6 Unsure

This reveals two key considerations for this paper. First, virtual schooling was offered in competition with other options, so it is not clear what type of institution is offering the virtual instruction. This could include students in public schools just switching to remote as well as actual school type switches (ex. Privately offered remote instruction). For our regression models, we include virtual schooling in public schooling but for alluvial diagrams we maintain the distinction. Additional details can be found in our **Limitations** section. Secondly, families were able to explicitly report that their children were not enrolled and not homeschooling. This is an advantage compared to administrative data and leaves little room for interpretation for non-enrolled students. For analyses we combine charter, magnet and virtual with public and non-enrollment with Other.

Sample Characteristics

Demographic Variables

Demographic variables provide essential context for understanding school choice decisions during the pandemic. This section summarizes key characteristics of the sample, including respondent gender, age, race, income, education, and urbanicity. The full set of descriptive statistics can be found in **Appendix C**.

Around 36% of the observations came from male respondents, while 67% indicated that the respondent was married or living with a partner. Additionally, about 71% of the observations were from working individuals, with the number of working respondents increasing by nearly 5% from Phase 2 to Phase 4. In terms of race and ethnicity, approximately 60% of the sample identified as White, followed by 23% Hispanic/Latino, 10% Black, and 7% Asian or other. Geographically, 18% of respondents were from rural areas, 34% from urban areas, and 48% from suburban (mixed) areas. Respondents were also asked about immunocompromising medical conditions, with 48% indicating they were immunocompromised, as captured by an indicator variable. Education and income levels are key to understanding the respondents' socioeconomic circumstances: 43% of the sample were college graduates, while 22% had a high school education or less. Income distribution was fairly even, with approximately 36% in the low-income group earning less than $40,000, 32% in the middle-income group earning between $40,000 and $100,000, and 32% in the high-income group earning more than $100,000 annually.

Perceived Risk, Institutional Trust, Media Trust, Fox News and Political Leanings

The following variables were included to assess attitudes surrounding the pandemic and political perceptions. A full breakdown of these variables can be found in **Appendix B.**

Factor analysis was used to generate a perceived risk of COVID-19 variable, using responses regarding their estimated chance of contracting and dying from COVID-19. Questions regarding the perceived risk of COVID-19 naturally aligned with the UCA survey waves, as they focused on pandemic-related attitudes during earlier phases. Therefore, perceived risk is excluded from the analysis in phases 3 and 4.

Respondents were asked about their level of trust in various media sources like ABC, NBC and local TV news. These responses were summarized through another factor variable. Upon comprising this variable, we noticed the uniqueness of Fox News responses and thus separated it from the factor index as its own variable.

Similarly, a factor analysis was created for trust in institutions by retrieving respondents' assessments of trustworthiness of various health organizations like the CDC (Center for Disease Control), the US Department of Health and Human Services, and local public health officials.
Lastly, we derive political leanings from UAs election polls and post-election surveys that were carried out from August 11, 2020, to November 2, 2020. We were able to retrieve political and urbanicity data for around 76% of observations. We summarized this voting behavior through an indicator variable where those who voted or intended to vote for Trump are assigned a value of 1. About 40% of observations were from respondents that either reported intending to or voting for Trump.

These demographic, socioeconomic, and attitudinal variables provide critical context for understanding the factors influencing school choice decisions throughout the pandemic.

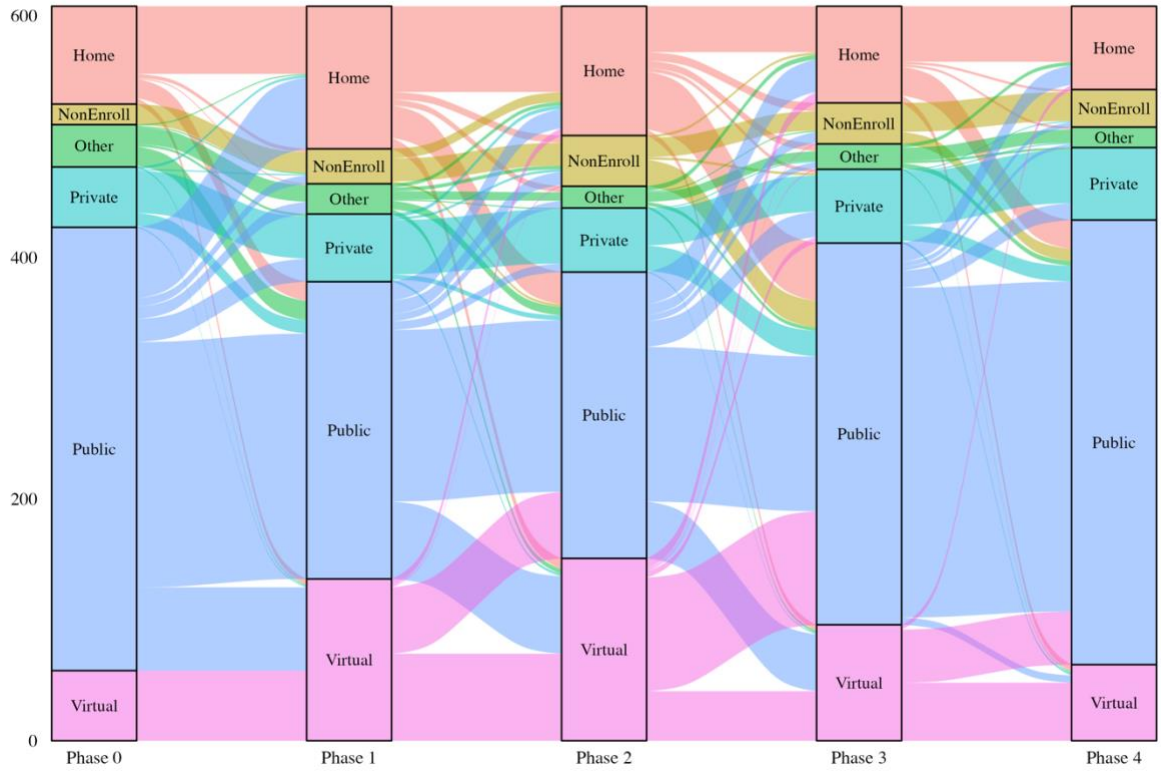
School Mode Overview

To better understand the transitions families made throughout the pandemic, we divide survey waves by phase and analyze the behavior of "switchers"—students who experienced a change in their schooling mode during the pandemic. Additionally, we split Phase 1 into Phase 0 (April to August 2020) and Phase 1 (September to December 2020) to visualize switches from the initial shock of the pandemic in the school year 2019-2020. The figure below corroborates findings from previous literature, which highlight significant shifts away from public schools during the 2020-21 school year (Phase 1-2; Census Bureau Pulse Survey, 2021; Burtis & Goulas, 2023; Dee, 2023). Notably, nearly 2 percent of students moved to virtual schooling in Phase 1 marking it as the most substantial movement during this period. This accounts for close to 34 percent of total switches for this phase. It is important to mention that respondents were asked to choose between virtual and other options like public, private, homeschooling and non-enrollment. This means if families chose virtual, we are not sure who was offering the instruction

so this could be under or overestimating school type switches. Switches to homeschooling accounted for the second biggest movement (~1.6% of total students). Private schooling received less switchers but still accounted for ~0.5% of students total in Phase 1. Non-enrollment peaked at Phase 2 receiving about 9 percent of switchers and ~0.6% of total students. Notably, non-enrollment grew by almost 60% in our sample from the beginning of the pandemic to Phase 2 and ~33% from beginning to end. It isn't until Phase 3 where we see significant movements back to public schooling (~5% of total students, ~54% of switchers). Interestingly, public school enrollment in our sample appears to have rebounded by Phase 4.

Figure 1: School Mode Overview

**Excluding non-switchers



Public School Exits

Approximately 3,115 respondents reported public school enrollment for their child in their first observation. Of these, about 18% transitioned their children to private, homeschooling, or other forms of schooling during the pandemic. These transitions reveal notable racial disparities. The proportion of Black students switching out of public schools (~19%) was roughly a third higher than that of White

students (~14%). Hispanic or Latino students exhibited switching rates similar to Black students, while Asian and other racial groups experienced the lowest proportion of exits (~10%). While about half of exits happen in the first phase for Black students, Hispanic/Latino students experience ~ 36% of their transitions in the first phase. Given the limited number of public school exits in later waves, our analysis of public school transitions primarily focuses on phases 1 and 2. A full breakdown of public exits by phase and type can be found in **Appendix E.**

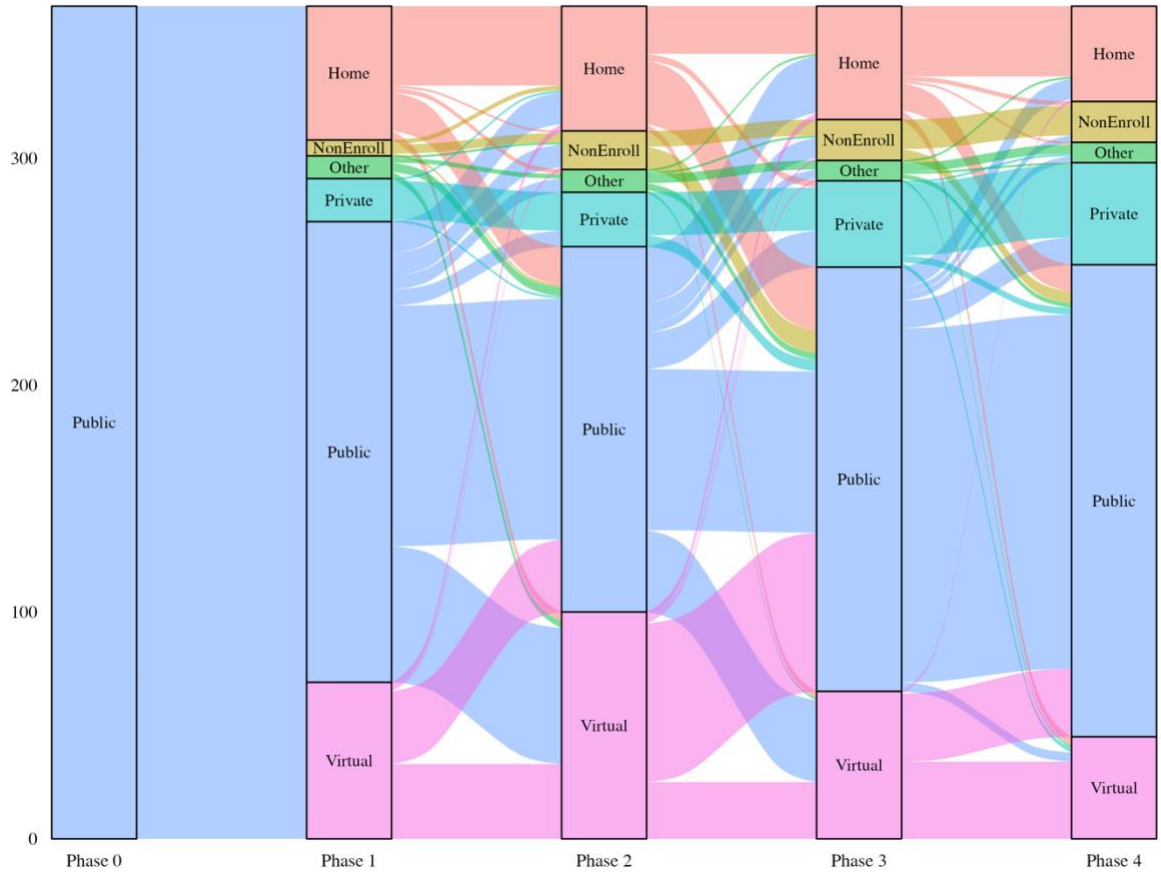
In addition to racial imbalances, income and education disparities were also evident. Interestingly, middle-income students see the highest proportion of transitions in Phases 1, low income students in Phase 2, while high income groups experience this in Phases 3 and 4. Still, absolute transitions were highest in Phase 1 for all groups. Approximately half of transitions for respondents with no college take place in Phase 1. In Phase 2, the highest proportion of switches is for those with some college experience and for Phase 3, this group is those with a 4-year college degree.

When analyzing the types of transitions out of public schools, approximately 56% were to homeschooling, ~18% were to private schooling, and the remaining to other schooling types. Significant racial/ethnic, income, and education disparities were observed in these patterns. The highest number of transitions to private schooling occurred among White students, high income households, and respondents with a 4-year college degree. Still, transitions to private schooling were very similar for all income groups. Transitions for homeschooling were most frequent among White students, low-income students, and respondents with some college. Most frequent transitions to Other schooling were similar except it was more common for respondents with a high school degree or less. Across all groups, transitions into homeschooling were more frequent than transitions into private schools.

To examine the shifts among public school starters, we plotted their school modes by phase. Notably, about 43% of the public-school starters in our sample had not returned to public schools by the end of the pandemic. Many students contributing to the recovery of public-school enrollment had initially started in other school modes during the pandemic. However, this recovery figure may be inflated, as some of these students may have originally been public school attendees prior to the pandemic but switched modes before they responded to UCA surveys.

Figure 2: Public School Starters

**Excluding non-switchers



Public School Entrances

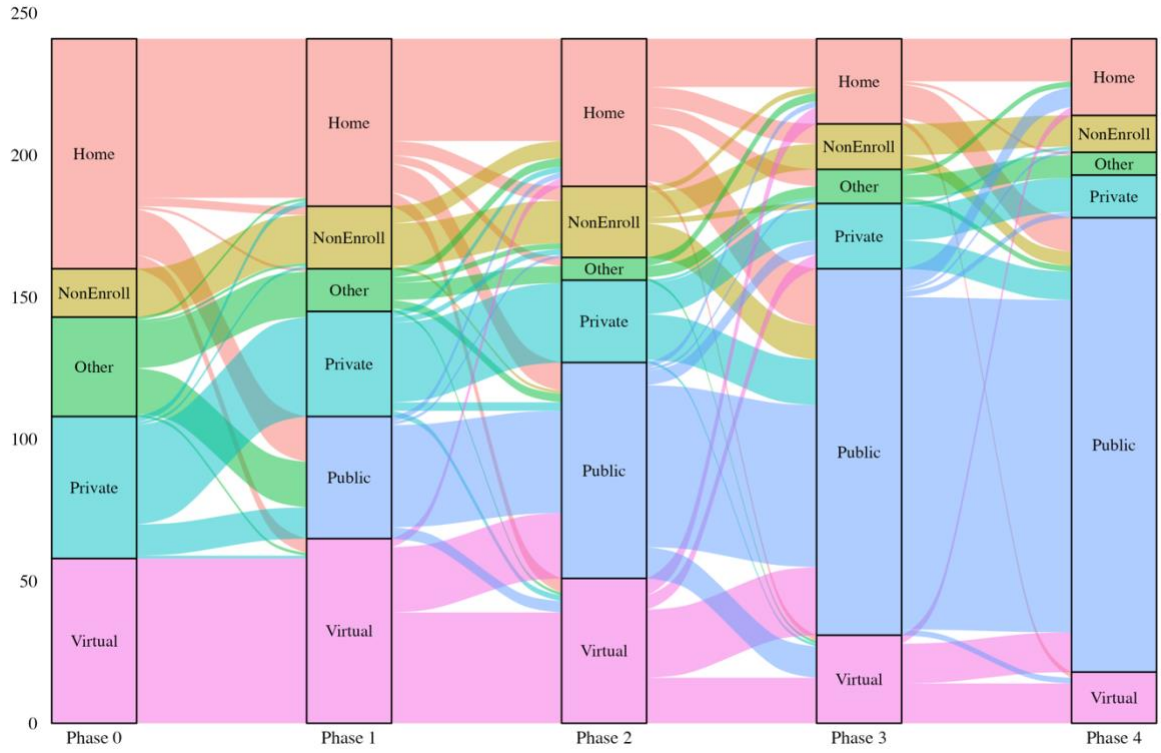
Transitions into public schooling include public school returners and those who we observe in public schooling for the first time in our sample. Racial differences in these transitions are more pronounced than those seen in public school exits. White, Black and Asian/Other students display similar entrance patterns where the highest levels are in Phase 1 and Phase 2 and later phases see less entrances. Even though Hispanic/Latino students entrances increase in Phase 3, the timing of these switches as a proportion of each race/ethnicities' total entrances is similar for all groups. Entrance numbers also vary by income, with low-income students showing the highest transition rates, followed by middle- and high-income students. While low-income students experience most of their transitions during Phase 1, other income groups have more consistent or even increasing entrances. Entrances by level of education followed similar patterns throughout all phases (Peaking in Phase 1 and dropping off slowly). A full breakdown can be found in **Appendix E**.

Transitions into public schools are most common among students switching from homeschooling followed by other and private. Black and Hispanic/Latino respondents had the highest rate of transitioning from homeschooling as a proportion of total returns (~60%). White and Asian/other students experienced a similar proportion of transitions being from homeschooling (~ 48%). Also, transitions from private schools were less common for Hispanic/Latino and Black students, with White students exhibiting the highest rates of such transitions. The patterns of transitions by income mirror those seen in public school exits: high-income households were most likely to transition from private schools, while low-income households had the highest rate of transitions from homeschooling. Finally, while transitions from other schooling types were relatively consistent across education levels, private school and homeschooling transitions showed more variation. More respondents transitioning from private school had higher levels of education while lower levels of education were more common for switches from homeschooling.

In **Figure 3**, we show transitions for those who switched at some point in our observations and whose first observation was NOT public school. Again, it is important to note that many of these respondents could have switched out of public schools before responding to UCA surveys so they would be more accurately described as public-school returners rather than students new to public schools. Additionally, virtual schooling could have been interpreted as public, private, or homeschooling causing an under or overestimation of school type transitions. Homeschooling lost about 66% of students while private schooling lost 70% of students from Phase 0 to Phase 4 in our sample. We see a slight increase in virtual school enrollment in Phase 1, but it continually declines throughout the rest of the pandemic.

Figure 3: Non-Public School Starters

**Excluding non-switchers



Econometric models and variables

We use logistic regressions to identify characteristics associated with the probability of switching school types. All results are presented as average marginal effects. We build this model sequentially beginning with only basic demographic characteristics (set A). Set B adds variables related to perceived risk and trust in various institutions to capture general attitudes toward COVID-19. Finally, a voting variable is included in set C to control for underlying political motivations.

A:

$$x'_j = \{raceEthnicity_i, income_i, education_i, age_i, gender_i, maritalStatus_i\ \ ageOfChild_i, urbanicity_i, numberOfChildren_{i,} surveySource_i, working_i\}$$

B:

$$x'_j = \{raceEthnicity_i, income_i, education_i, age_i, gender_i, maritalStatus_i,\ ageOfChild_i, urbanicity_i, numberOfChildren_{i,} surveySource_i, working_i, perceivedRisk_i, mediaTrust_i, institutionalTrust_i\}$$

C:

$$x'_j = \{raceEthnicity_i, income_i, education_i, age_i, gender_i, maritalStatus_i, ageOfChild_i, urbanicity_i, numberOfChildren_{i,} surveySource_i, working_i, perceivedRisk_i, mediaTrust_i, institutionalTrust_i, trumpVoter_i\}$$

Logistic models quantify the relationships between school type choices and covariates, assuming a linear relationship. To check robustness, we also apply a more flexible model, Random Forests, to see if the relationships identified in overall multinomial analyses are consistent across both methods.

Overall

The first analysis is a multinomial regression where the outcome variable is school type. We split the analysis into four phases to better understand how the importance of factors influencing school choice changes over time. Public school is the reference category, with Private, Homeschooling, and Other represented as the alternative outcomes. Controls for the survey wave are included.

$$P(Y_i = j|x_i) = \frac{exp(\beta'_j x_i)}{\sum_{l=1}^{4} exp(\beta'_j x_i)}, j = \begin{cases} 1\ for\ Public \\ 2\ for\ Private \\ 3\ for\ Homeschool \\ 4\ Other \end{cases}$$

Discrete Choice Duration Models

Public Exit by Phase

The second analysis focuses only on public-school starters. Using a logit regression, the outcome variable is coded as 0 for pre-switching observations and 1 for switches to private, homeschooling, or other types of schooling. After an individual switches out of public school, they are excluded from the analysis. This model examines the characteristics that influence how long public-school starters "survive" in the public school system. We run several binary models by phase.

$$P(Y_i = j|x_i) = \frac{\exp\left(\beta'_j x_i\right)}{\sum_{l=1}^{2} \exp\left(\beta'_j x_i\right)}, j = \begin{cases} 1\ for\ switch\ out\ of\ public\ school \\ 0\ before\ the\ switch \end{cases}$$

Public Exit by Type

Similar to public exit by phase, we only include public school starters. The main difference is that we make a distinction by what type of schooling the respondent exited to. Respondents who reported attending private school, homeschooling, or other school types after having reported public school were

denoted as 1, 2, or 3. For those who never exited public school, all observations remained 0. Observations after "the switch" were dropped. This includes all phases into a single multinomial regression.

Public Return by Phase

The public return analyses include only those that did not have public school as their first observation or who exited and then came back to public schools. 0 represents observations prior to switching into public school, while 1 represents a switch from private, homeschooling, or other types of schooling. This model aims to examine the determinants of those who either return to or enter public school for the first time. Observations after "the switch" are dropped.

$$P(Y_i = j|x_i) = \frac{\exp\left(\beta'_j x_i\right)}{1 + \exp\left(\beta'_j x_i\right)}, j = \begin{cases} 1\ for\ switch\ into\ public\ school \\ 0\ before\ the\ switch \end{cases}$$

Public Return by Type

Again, we make a distinction by what type of schooling the respondent returned from. Before switching to public school, respondents who reported attending private school, homeschooling, or other school types were denoted as 0. For those who never entered public school, all observations remained 0. The first observation indicating entry into public school was coded as 1 for private school, 2 for homeschool, and 3 for other depending on their previous observation. For respondents who began in public school, exited, and then returned, we drop their initial public-school observations. This includes all phases into a single multinomial regression and observations after this entrance are dropped.

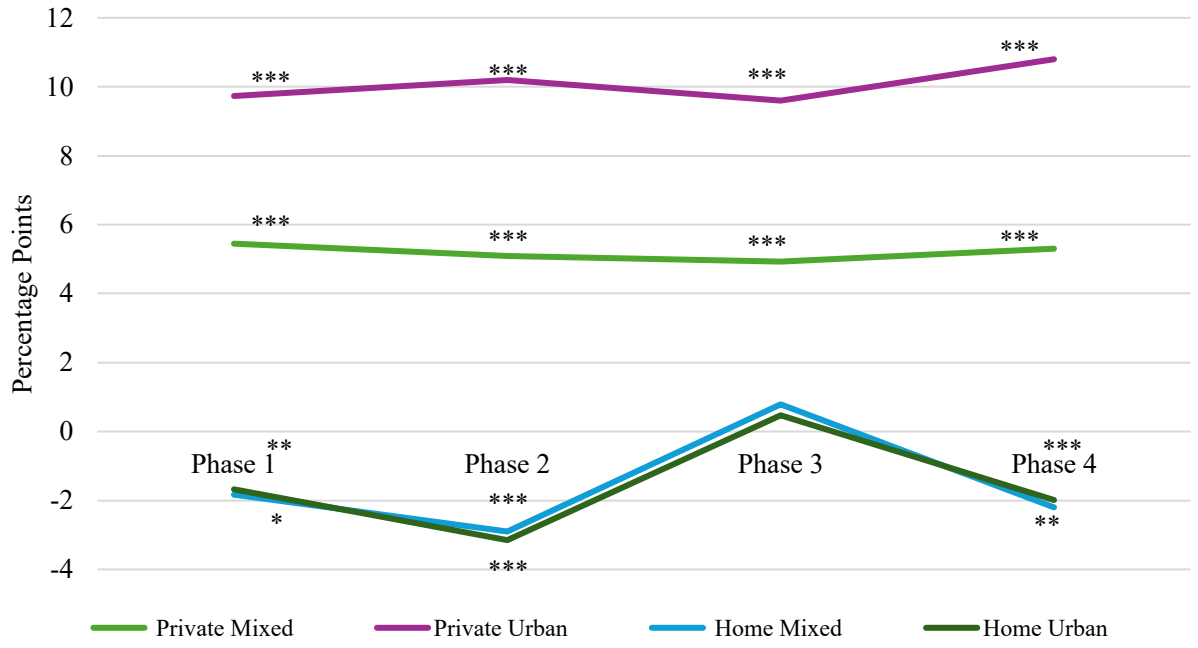
$$P(Y_i = j|x_i) = \frac{exp(\beta'_j x_i)}{\sum_{l=1}^{4} exp(\beta'_j x_i)}, j = \begin{cases} 0\ for\ Private, Homeschool, Other \\ 1\ switch\ to\ Public\ from\ Private \\ 2\ switch\ to\ Public\ from\ Homeschool \\ 3\ Switch\ to\ Public\ from\ Other \end{cases}$$

Results

Multinomial by Phase

We begin with the simpler model including only demographic characteristics to draw associations with school type. Due to the low number of survey waves attributed to Phase 3, only the basic model including demographic characteristics converges for all phases. A full set of results can be found in **Appendix F.** The strongest associations are found with urbanicity, income level, working status, education, age of child, and race/ethnicity. Private schooling was highly associated with more urban environments compared to public school. As seen in **Figure 4** below, families living in urban environments were 10 percentage points more likely to report private schooling compared to rural environments. Homeschooling hovered around 2 negative percentage points for families living in urban and suburban (mixed) environments. Although they see a reduction in magnitude after adding trust factors and political leanings, urbanicity remains a strong predictor.

Figure 4: Urbanicity by School Type

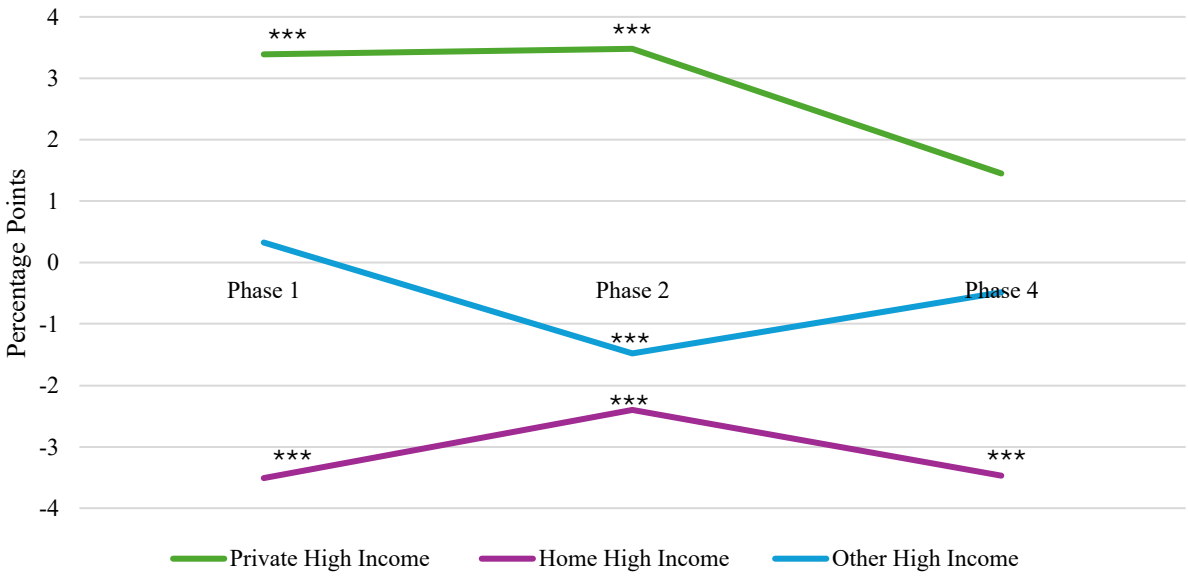


*** p<0.01, ** p<0.05, * p<0.1; Marginal Effects as percentage points using Set A

Income levels and working status were also associated with different school types. Across Set A, B and C, the magnitude and significance of these marginal effects are consistent. Working status was found to

be positively associated with private schooling by about 2.5 percentage points with the very last phase losing significance and magnitude. Homeschooling shows stronger but negative associations (3.5 percentage points in Phase 2) with working status as compared to public school attenders. Those in other schooling more closely resembled those in public school in terms of income during the pandemic. Again, like associations with working status and urbanicity, private and homeschooling attenders have opposite associations with income. **Figure 5** shows the varying association with high income households compared to those in the lower income bracket. Interestingly, by Phase 4, homeschooling negative associations with higher income maintain significance while associations with private and other school type attenders lose it.

Figure 5: Associated Income by School Type



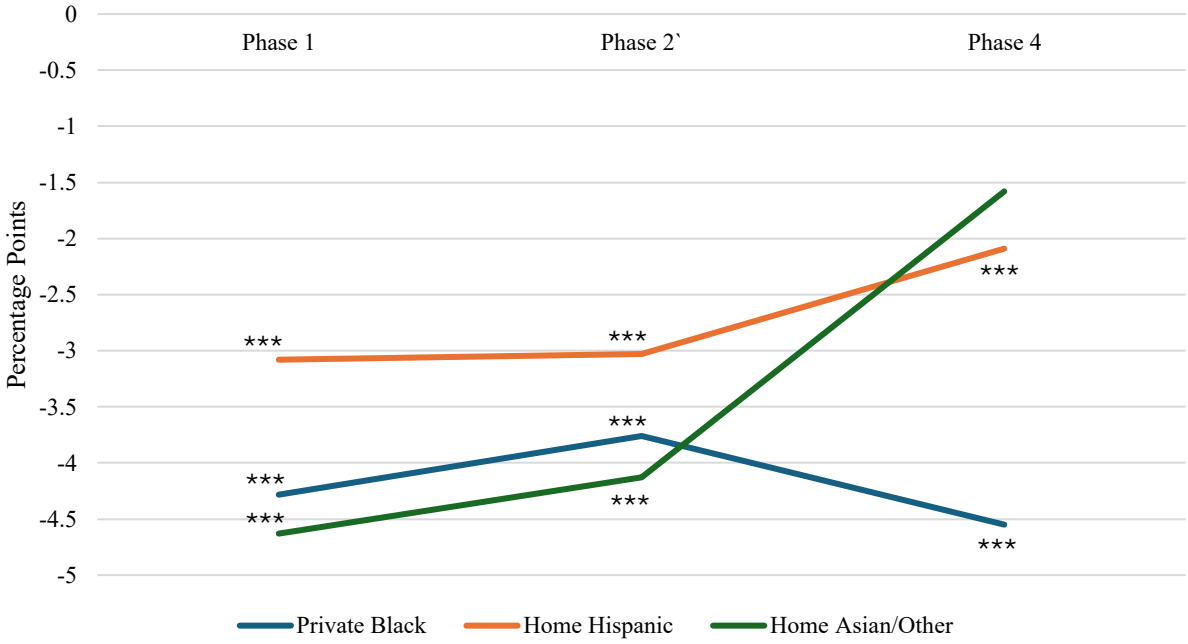
*** p<0.01, ** p<0.05, * p<0.1; Marginal Effects as percentage points using Set C

Even after including trust factors and political leanings, we find that private schooling was positively associated with 4-year college degree holders. Staying significant throughout, college degree earners were up to ~6 percentage points (Phase 4, Set C) more likely to have their students in private schooling. Yet we find that other associations between education and school type were not significant in these models.

Nearly all models found negative associations between racial and ethnic minorities and private, home and other schooling, ceteris paribus. After including trust factors and political leanings, there was varying changes to the associated effects. During Phase 1, the full model reduced racial/ethnic effects by a about 0.04 percentage points at the maximum (Black). During Phase 4, Set B had ~0.04 percentage points lower for homeschooled Hispanic families than Set A. Private schooling was negatively associated with Black

families ranging from ~4.5 to 1.5 percentage points throughout. Additionally, Hispanic families were from ~2-3 percentage points less likely to pursue homeschooling compared to White families throughout. It is important to consider that the UAS survey waves allowed respondents to claim their child was attending "virtual" school. These responses were categorized into public schooling but could be including students who were attending virtual homeschooling or private programs.

Figure 6: Race/Ethnicity Associated by School Type (Set C)



*** p<0.01, ** p<0.05, * p<0.1

As supported by previous literature, the age of the child was highly associated with certain school types. Compared to Pre-K and kindergarten students, high school students were about 3.5 percentage points less likely to attend private school and 3.6 percentage points less likely to be homeschooled in Phase 2. The magnitude of these effects grows over the course of the pandemic reaching -6.0 percentage points for high schoolers attending private school. This increase could be more reflective of parents pulling younger students out of public schooling during the pandemic compared to high school students.

Now we examine marginal effects associated with trust factors and political leanings. Media trust was significantly positively associated with both home and other schooling in Phase 2 but with small marginal effects of ~1% and ~0.6% respectively. On the contrary, institutional trust was negatively associated with home and other schooling throughout. At its peak in Phase 2, institutional trust was associated with

homeschooling ~2 less percentage points compared to public schooling. Trust in Fox News and political associations were significant for private schooling throughout. Being a trump voter was associated with about 2 percentage points higher likelihood of private schooling in Phase 1 and this effect went up to ~3 percentage points in Phase 2. Trust in Fox News associations hover around 1.6 percentage points for the likelihood of attending private school. Lastly, trust in Fox News showed positive associations with homeschooling as well. However, these associations with homeschooling did not match the magnitude of private schooling and are no longer significant at Phase 4.

Factors to match in direction for private and homeschoolers and be significant were race/ethnicity, age of child, trust in Fox News, and voting for Trump.

Public School Exits by Phase

The focus of these models is to understand the changes or factors that are associated with switching out for the subset of public-school starters. For this analysis we focus on results from Phase 1 and 2. The full set of results can be found in **Appendix G.** The most explanatory factors were income, trust in Fox News and institutional trust. Being in the highest income bracket was associated with a 0.88 percentage points decrease in probability of exiting public schools in the overall public exit analysis. This was strongest in Phase 2 using the full set of covariates reaching 0.93 negative percentage points. While income associations became stronger, trust in Fox News associations became slightly weaker from Phase 1 to 2. Yet, trust in Fox News was also significant throughout. In Phase 1, increasing by 1 point in trust was associated with a 0.90 percentage points higher likelihood of exiting public schools holding all else equal (Set C). For Phase 2 analyses, the marginal effects hovered around 0.5 positive percentage points. Lastly institutional trust was negatively associated with switching out of public school for public school starters. However, this association was only significant in Phase 2 analyses. In the full set, the association was of about 0.69 percentage points. Again, it is important to consider only 1.89% of observations in Phase 1 and 0.92% of observations in Phase 2 indicate an exit from the public sector.

Public School Entrances by Phase

Unlike the analysis measuring public school exiters described above, in this case, we include the full sample and use a dummy of when families switch into public school as the outcome variable. This includes both public school returners and those that entered public school for the first time in our observations. The full set of results can be found in **Appendix H.** The overall model (including all phases) finds significant associations at the 99% level for education, age of child, and political leanings. Higher levels of education are negatively associated with returning to public school throughout at almost

7 negative percentage points for 4-year college degree holders. The student being in middle or junior high school was positively associated with returning by ~ 5 percentage points. On the contrary, identifying as a Trump voter was negatively associated with returning to public schools by about 5 percentage points. Interestingly, income is a significant factor in Phase 1 analyses with the magnitude reaching 13.1 negative percentage points after including all covariates (Set C). Some racial/ethnic effects were found to be significant in Phase 4 where Black students were ~ 24 percentage points more likely to return compared to white students. Given that by phase from 1 to 4, 11.32, 6.52, 17.51, and 10.47 percent of observations indicated an entrance into public schools, even effects as low as 5 percentage points are meaningful.

Public Exits by Type of Switch

Since previous literature has pointed to substantial differences between homeschooling and private school switchers, we include a survival model that accounts for type of switch and a general multinomial model split up by phase. Unfortunately sorting by both timeframe and type of switch simultaneously divides the data into overly small subsets, limiting reliability of results. Therefore, we regress school type on the full set of covariates over all 4 phases. Perceived risk and exposure to remote schooling is not included in this analysis as these variables are only available for early survey waves. This sample is restricted to public school starters. The full set of results can be found in **Appendix I.**

Racial effects, education, age of child and trust in Fox News were found to be significant at the 95% level for private schooling. Black students and Asian and other ethnicities were almost 0.2 percentage points less likely to exit to private schools. Greater racial effects were found for Asian and other for other schooling types where these students had an almost 3 percentage point negative association with exiting. Being a college degree holder was positively associated with exiting to private schools by ~0.2 percentage points. This is stronger and significant at the 99% level for homeschooling at ~0.5 percentage points. The stronger association could be due to a parent with sufficient education required to assist in their child's homeschooling. Older students were negatively associated with private schooling holding all else equal. Yet, exiting to other schooling types was significant for high school students by about 3.8 positive percentage points. Associations with income for exiting into private school were not found to be significant. Being in the highest income bracket was negatively associated with homeschooling and other types of schooling by about ~0.6 and ~0.4 percentage points respectively. Trust in Fox News was positively associated with exiting to private, home and other schooling types. The strongest of these effects was found to be for homeschooling although they all hover around 0.15 percentage points.

Public Returns by Type of Switch

This outcome variable denotes a switch into public school making a distinction by where they were last observed. Again, perceived risk and exposure to remote options were removed as these variables do not span the entirety of the survey waves. The full set of results can be found in **Appendix J.** The most important factors for were education, urbanicity, and political leanings. Being a college degree holder was negatively associated with returning from homeschooling by ~ 5.4 percentage points. Urbanicity was positively associated with returning from private schooling by ~ 3.4 percentage points. It is important to consider that urban environments were more associated with exiting public schools in favor of private schooling in the first place. On the contrary, urbanicity was negatively associated with returning to public schools from other types of schooling by about 2.4 percentage points. The student being in middle or junior high school was found to be positively associated with returning from homeschooling by ~3.5 percentage points but not significant for other types of returns. Lastly, being a trump voter was found to be negatively associated with returning to public schools from home or other types of schooling. These associations were stronger for home schooling and significant at the 99% level (3.9 percentage points). Returns for other school types were found to be significant at the 90% level; their association with being a Trump voter was 2 percentage points less than home schooling.

Limitations

Throughout the UCA survey waves, there were inconsistencies in the way families were asked about their child's school type. Spanning throughout almost all survey waves (UAS 272 forward), the survey provides "Virtual School" as an option separate from public, private and homeschooling. To focus on public school movements, "Virtual school" was categorized as public although families might have interpreted this as public remote or private/home virtual school. We suspect this adds noise especially for the multinomial school type analysis containing non-enrolled students as well as students who would have otherwise been sorted as public, private, or homeschooled.

Secondly, families are categorized into either White only, Black Non-Hispanic, Asian/Other Non-Hispanic, and Hispanic/Latino. Due to the low number of observations for Pacific Islander and Native American groups, these families are combined with Asian families. This could mask some effects these groups may be experiencing given previous literature notes Asian school choice movements are more mirrored by white families than by other minority groups. Additionally, some respondents categorized themselves as being part of multiple racial or ethnic groups. The intersectionality of these effects may also be masked by this categorization.

Conclusion

As public schools struggled to retain students, private and homeschooling enrollment grew during the COVID-19 pandemic. By October 2020, close to 1.6% of students in our sample had transitioned into homeschooling; private schooling received less switchers but still accounted for 0.5% of students. Non-enrollment grew by almost 60% in our sample from the beginning of the pandemic to the end of December 2020 and ~33% from April 2020 to February 2024. From Phase 2 to 3, more than half of the switches were into public schools. By Phase 4, we observe almost a full recovery of enrollment for public schools. Still, these enrollment numbers are far from the pre-pandemic growth rates that public schools were experiencing. Given enrollment's direct tie to revenue, these losses can materialize into school closings, personnel layoffs and program cuts which are damaging to attending students. Additionally, identifying the types of families that spent time outside of public schools can clarify potential educational disparities even if these families returned. Some evidence points to differences in educational losses experienced by public versus private school attenders during the pandemic.

For this paper, we used multinomial regressions and duration models for switches into and out of public schools. These analyses reveal several key factors that influenced school choice decisions during the pandemic, which are important for policymakers and public-school districts to consider.

Despite public schools seeing significant recovery, the presence of socioeconomic and racial/ethnic factors points to some sorting that could differentiate educational outcomes. Although in differing directions, private (higher income) and homeschooling (lower income) decisions were significantly associated with income in almost all multinomial analyses. For the subset of public school starters, we found an association of nearly a full percentage point (0.93, Phase 2) between exiting public schools and being in the high-income bracket compared to the low-income bracket. Given that only 0.92% of observations in this phase were exits, this effect is substantial. We find in public exit by type analyses that for the subset of public school starters, these income associations are almost entirely driven by homeschool switches. In line with the Census Bureau Pulse Survey, 2021, racial and ethnic minorities were negatively associated with school types outside of public school. In the overall multinomial analysis, we observed particularly strong negative associations between Black families and private schools, as well as between Hispanic/Latino, Asian, and other families and homeschooling. Lastly, higher levels of education were found to be significantly negatively associated with public schools in almost all models. Public school exits into homeschooling were positively associated with having a 4-year degree by ~0.5 percentage points (0.71% of observations were into homeschooling overall). Similarly, having a college degree amounted to a ~0.2% percentage point increase in probability to switching to private schools on

average (0.23% of observations were into private schooling overall). Continuing to monitor this is crucial as this enrollment is directly tied to revenue and diversity in public schools.

Given trust in Fox News was consistently a strong factor for public school exits but not return or public-school entrances, Fox News could serve as a proxy for families that have exited the public school system and were most resistant to return, indicating a potential increased divide by political ideology. In our public exit analysis, an increase in one point in trust in Fox News was associated with a 0.90 increase probability in exiting public schools, holding all else equal (1.89% of observations were exits in Phase 1). Being a Trump voter was negatively associated with returning from homeschooling by ~3.9 percentage points, ceteris paribus. Associations between political and trust variables and school type decisions could point to ideological diversity concerns.

Additionally, we observe changing associations depending on the phase of the pandemic. Being in the higher income bracket was more explanatory for school type at the beginning of the pandemic. Potentially signifying higher income families' greater mobility and ability to respond to the pandemic. On the contrary, being in an urban environment maintained its magnitude of positive association with private schooling throughout our sample. Associations of leaving public schools by age of the child grow in later phases of the pandemic. As supported by Burtis & Goulas (2023), high school students were negatively associated with leaving public schools compared to Pre-K/Kindergarten students.

Future research could explore how various phases of the pandemic, and the different types of switches simultaneously interact to influence these explanatory factors. Additionally, the growth of non-enrollment observed in our sample could be cause for concern. Investigating how these non-enrollment decisions take place, and their long-term impacts is important to understanding the effect of the pandemic on education.

Policy Implications

Given that a large number of public-school students exited for homeschooling and non-enrollment modes, there are elevated concerns for regulation. Reporting for these modes varies from state to state and non-enrollment students would not be observed in administrative data. Losing track of these students makes it increasingly difficult to properly assess the educational system.

Although public schools have shown significant recovery, public schools should still reconsider their policies and strategies to retain the families most likely to leave. As school choice policies continue to be

implemented, public schools will need to compete with other school types in order to reduce losses in enrollment. Additionally, the variance between public school exiters could imply that public schools will need to try different strategies to retain different groups (i.e. Low income exiters to homeschool, high income exiters to private school).

We may need targeted policy to support students who have suffered from educational disparities during the pandemic. For example, even if students enrolled in public, private, or homeschooling programs after the pandemic, the time they spent as non-enrolled students could materialize into educational setbacks.

Acknowledgements

The project described in this paper relies on data from survey(s) administered by the Understanding America Study, which is maintained by the Center for Economic and Social Research (CESR) at the University of Southern California. The content of this paper is solely the responsibility of the authors and does not necessarily represent the official views of USC or UAS.

This research was supported by the University of Arkansas Honors College.

Research reported in this publication was supported by the National Institute on Aging of the National Institutes of Health and in part by the Social Security Administration under Award Number U01AG077280. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

References

Abdulkadiroglu, A., & Andersson, T. (2022). *SCHOOL CHOICE.* NATIONAL BUREAU OF ECONOMIC RESEARCH.

Bacher-Hicks, A., Musaddiq, T., Goodman, J., & Stange, K. (2024). *The stickiness of pandemic-driven disenrollment from public schools.* Economics of Education Review.

Burtis, E., & Goulas, S. (2023, October 12). *Declining school enrollment since the pandemic.* Retrieved from Brookings Institute: https://www.brookings.edu/articles/declining-school-enrollment-since-the-pandemic/#:~:text=We%20find%20that%20there%20were,schools%20experienced%20such%20a%20decline.

Calarco, J. M., Coleman, M., & Halpern-Manners, A. (2021). *Mechanisms of Stratification in In-Person Instruction in the Wake of COVID-19.* SocArXiv Papers.

Camp, A. M., & Zamarro, G. (2021). Determinants of Ethnic Differences in School Modality Choices During the COVID-19 Crisis. *American Educational Research Association*.

Camp, A., Johnson, A., & Zamarro, G. (2023). *Revisiting Ethnic Differences in In-Person Learning During the 2021-22 School Year* . Journal of School Choice.

Dee, T. S. (2023). *Where the Kids Went: Nonpublic Schooling and Demographic Change during the Pandemic Exodus from Public Schools.* Urban Institute.

Dee, T. S., & Murphy, M. (2021). *Patterns in the Pandemic Decline of Public School Enrollment.* Sage Journals.

Eggleston, C., & Fields, J. (2021). *Census Bureau's Household Pulse Survey Shows Significant Increase in Homeschooling Rates in Fall 2020.* Retrieved from U.S. Census: https://www.census.gov/library/stories/2021/03/homeschooling-on-the-rise-during-covid-19-pandemic.html

ERB. (2024, September 24). *Bouncing Back from COVID: Key Findings from ERB's 2023-2024 Assessment Data*. Retrieved from https://www.erblearn.org/blog/learning-loss-recovery-independent-schools/

Gordon, M. F., de la Torre, M., Cowhy, R. J., Moore, P. T., Sartain, L., & Knight, D. (2018). *School Closings in Chicago: Staff and Student Experiences and Academic Outcomes.* Chicago, IL: University of Chicago Consortium on School Research.

Grossmann, M., Reckhow, S., Strunk, K. O., & Turner, M. (2021). *All States Close but Red Districts Reopen: The Politics of In-Person Schooling During the COVID-19 Pandemic.* Retrieved from American Educational Research Association: https://doi.org/10.3102/0013189X211048840

Han, C., Raymond, M. E., Woodworth, J. L., Negassi, Y., Richardson, W. P., & Snow, W. (2017). *Lights off: Practice and impact of closing low-performing schools.* Stanford, CA: Research Report, Center for Research on Education Outcomes, Stanford University.

Institute of Education Statistics. (2021). *IES Monthly School Survey Dashboard.* U.S. Department of Education.

Kelly, L. B. (2023). *What Do So-Called Critical Race Theory Bans Say?* Retrieved from American Educational Research Association: https://journals.sagepub.com/doi/full/10.3102/0013189X231159382

McCluskey, N., & Ekins, E. (2024). Survey: 46 Percent of Private Schools See Enrollment Rise. *CATO Institute*.

Miles, K. (2021). *More Black families are homeschooling their children, citing the pandemic and racism.* Retrieved from NPR: https://www.npr.org/2021/12/13/1061787233/more-black-families-are-homeschooling-their-children-citing-the-pandemic-and-rac

Murnane, R. J., & Reardon, S. F. (2018). *Long-Term Trends in Private School Enrollments by Family Income.* American Educational Research Association.

National Center for Education Statistics. (2023). *Public School Enrollment.* U.S. Department of Education, Institute of Education Sciences.

Romano, S. D., Blackstock, A. J., Taylor, E. V., Felix, S. E., Adjei, S., Singleton, C.-M., . . . Boehmer, T. K. (2021). *Trends in Racial and Ethnic Disparities in COVID-19 Hospitalizations, by Region — United States, March–December 2020.* Retrieved from Center for Disease Control: https://www.cdc.gov/mmwr/volumes/70/wr/mm7015e2.htm#suggestedcitation

Shao, W., & Hao, F. (2020). *Confidence in political leaders can slant risk perceptions of COVID–19 in a highly polarized environment.* Retrieved from Social Science & Medicine: https://www.sciencedirect.com/science/article/pii/S0277953620304548?via%3Dihub

U.S. Government Accountability Office. (2022). *Student Population Has Significantly Diversified, but Many Schools Remain Divided Along Racial, Ethnic, and Economic Lines.* Committee on Education and Labor.

United States Census Bureau. (2023). *U.S. Population Trends Return to Pre-Pandemic Norms as More States Gain Population.*

Woo, A., Wolfe, R. L., Steiner, E. D., Doan, S., Lawrence, R. A., Berdie, L., . . . Schwartz, H. L. (2022). *Walking a Fine Line—Educators' Views on Politicized Topics in Schooling.* Retrieved from RAND: https://www.rand.org/pubs/research_reports/RRA1108-5.html

Appendix

Appendix A: Phases by Dates and UAS Survey Source

Phase	Dates	Survey Source
Phase 1	April 15, 2020 to October 27, 2020	UAS 240, UAS 242, UAS 264, UAS 270, UAS 272, UAS 274
Phase 2	December 23, 2020 to January 16, 2021	UAS 276, UAS 278, UAS 280, UAS 282, UAS 340, UAS 342, UAS 344, UAS 346
Phase 3	September 23, 2021 to May 12, 2022	UAS 350, UAS 351, UAS 461
Phase 4	July 21, 2022 to February 04, 2024	UAS 475, UAS 502, UAS 559, UAS 584

Appendix B: Construction of Variables

Number of children in the household was derived counting all reported household members under the age of 18. Age of respondent was reported as a dummy of 5 age bins (Under 30, 30-39, 40-49, 50-59, 60 and over). An indicator of gender of the respondent was included as directly reported in surveys. Number of children in the household was derived counting all reported household members under the age of 18. Age of respondent was reported as a dummy of 5 age bins (Under 30, 30-39, 40-49, 50-59, 60 and over). Working includes part-time, full-time, and any work for pay. The UCA includes data on respondent's race within six categories (white, Black, American Indian or Alaskan Native, Asian, Hawaiian or Pacific Islander and mixed). It also records whether the respondent identifies as Hispanic or Latino. This was summarized through indicators for White, Black, Hispanic/Latino, and Asian/Other. Levels of education were separated by high school or lower, some post-secondary experience with no earned degree, and post-secondary degree and above. Similarly, we include three household income brackets: less than $40,000, $40,000-$100,000, and more than $100,000 per year. Urbanicity was merged from the UAS U.S. Election Longitudinal survey. Respondents are categorized as rural, mixed or in an urban environment.

Perceived Risk, Institutional Trust, Media Trust, and Fox News

To comprise the perceived risk of COVID-19 variable, a factor analysis was conducted using the responses to the following questions:

What is the chance (0-100) that you will get coronavirus within the next 3 months?

What is chance (0-100) that you will die from coronavirus given you contract it?

Perceived Risk Variable	**Factor**	**Uniqueness**
Perceived Risk of Infection from COVID-19	0.8693	0.2444
Perceived Risk of Death from COVID-19	0.8693	0.2444

Institutional Trust Variable	**Factor**	**Uniqueness**
Center for Disease Control	0.9035	0.1837
United States Department of Health and Human Services	0.8940	0.2007
Trusts Physician	0.6974	0.5136
Local Public Health Officials	0.8591	0.2620
World Health Organization	0.8554	0.2683

Similarly, a measure of media trust was derived from respondents' assessments of trustworthiness of CNN, MSNBC, NBC, CBS, ABC, and national newspapers. Fox News trust exhibited a significantly higher uniqueness value during the factor analysis and was therefore separated into its variable.

Media Trust Variable	**Factor**	**Uniqueness**
ABC	0.9341	0.1275
CBS	0.9363	0.1234
CNN	0.8917	0.2049
Local newspaper	0.8238	0.3214
Local TV news	0.8339	0.3046
MSNBC	0.9011	0.1879
National newspaper	0.8942	0.2004
NBC	0.9388	0.1186
Public TV or radio	0.8519	0.2743

Exposure to Remote Learning Variable

Given the variance in school reopening policies in Phase 1, we include a measure of remote learning supply in early analyses. This variable was derived making use of phone survey data from districts collected by MCH. Specifically, we use MCH's Strategic Data as reported in October 2020 which accounts for 78% of districts nationwide. Remote learning exposure estimates are constructed at the census tract level as a proportion of students whose districts only offered remote learning. In other words, this measure represents that absolute unavailability of in-person learning options (including hybrid).

Appendix C: Descriptive Statistics

	Phase				
	1	2	3	4	Total
N	6,260	12,768	5,294	5,947	30,269
	(20.7%)	(42.2%)	(17.5%)	(19.6%)	(100.0%)
Gender	0.358	0.361	0.354	0.345	0.356
	(0.479)	(0.480)	(0.478)	(0.475)	(0.479)
Race and Ethnicity					
White	3,821	7,622	3,239	3,425	18,107
	(61.1%)	(59.7%)	(61.2%)	(57.6%)	(59.9%)
Black	629	1,262	508	675	3,074
	(10.1%)	(9.9%)	(9.6%)	(11.4%)	(10.2%)
Hispanic/Latino	1,376	2,947	1,189	1,406	6,918
	(22.0%)	(23.1%)	(22.5%)	(23.6%)	(22.9%)
Asian/Other	430	929	355	440	2,154
	(6.9%)	(7.3%)	(6.7%)	(7.4%)	(7.1%)
Marital Status	0.687	0.677	0.700 (0.458)	0.691	0.686
	(0.464)	(0.468)		(0.462)	(0.464)
Education					
No College	1,417	2,813	1,118	1,283	6,631
	(22.7%)	(22.0%)	(21.1%)	(21.6%)	(21.9%)
Some College	2,196	4,536	1,886	2,028	10,646
	(35.1%)	(35.5%)	(35.6%)	(34.1%)	(35.2%)
College Graduate	2,643	5,412	2,287	2,633	12,975
	(42.2%)	(42.4%)	(43.2%)	(44.3%)	(42.9%)
Household Income					
Low Income	2,345	4,853	1,827	1,964	10,989
	(37.5%)	(38.0%)	(34.5%)	(33.1%)	(36.3%)
Middle Income	2,101	4,114	1,697	1,820	9,732
	(33.6%)	(32.2%)	(32.1%)	(30.6%)	(32.2%)
High Income	1,811	3,797	1,768	2,157	9,533
	(28.9%)	(29.7%)	(33.4%)	(36.3%)	(31.5%)

Age of Respondent					
Under 30	324	733	276	326	1,659
	(5.2%)	(5.7%)	(5.2%)	(5.5%)	(5.5%)
30 to 39	1,433	2,880	1,349	1,635	7,297
	(22.9%)	(22.6%)	(25.5%)	(27.5%)	(24.1%)
40 to 49	2,479	4,926	2,184	2,489	12,078
	(39.6%)	(38.6%)	(41.3%)	(41.9%)	(39.9%)
50 to 59	1,407	2,891	1,051	1,078	6,427
	(22.5%)	(22.7%)	(19.9%)	(18.1%)	(21.2%)
60 and over	613	1,328	434	418	2,793
	(9.8%)	(10.4%)	(8.2%)	(7.0%)	(9.2%)
Number of Kids					
1	3,080	6,497	2,517	2,865	14,959
	(49.2%)	(50.9%)	(47.5%)	(48.2%)	(49.4%)
2	1,845	3,714	1,653	1,843	9,055
	(29.5%)	(29.1%)	(31.2%)	(31.0%)	(29.9%)
3	935	1,813	747	835	4,330
	(14.9%)	(14.2%)	(14.1%)	(14.0%)	(14.3%)
4	267	486	247	268	1,268
	(4.3%)	(3.8%)	(4.7%)	(4.5%)	(4.2%)
5	79	148	71	78	376
	(1.3%)	(1.2%)	(1.3%)	(1.3%)	(1.2%)
6	36	71	33	30	170
	(0.6%)	(0.6%)	(0.6%)	(0.5%)	(0.6%)
7	15	34	19	20	88
	(0.2%)	(0.3%)	(0.4%)	(0.3%)	(0.3%)
8	0	0	4	6	10
	(0.0%)	(0.0%)	(0.1%)	(0.1%)	(0.0%)
9	0	0	0	2	2
	(0.0%)	(0.0%)	(0.0%)	(0.0%)	(0.0%)
10	3	5	3	0	11
	(0.0%)	(0.0%)	(0.1%)	(0.0%)	(0.0%)
Urbanicity					
Rural	1,024	1,840	656	610	4,130
	(18.8%)	(17.5%)	(17.3%)	(17.2%)	(17.7%)

Mixed	2,624	5,020	1,853	1,701	11,198
	(48.1%)	(47.7%)	(48.8%)	(47.8%)	(48.0%)
Urban	1,812	3,664	1,286	1,245	8,007
	(33.2%)	(34.8%)	(33.9%)	(35.0%)	(34.3%)
Medical Condition	0.483	0.481	0.475	0.478	0.480
	(0.500)	(0.500)	(0.499)	(0.500)	(0.500)
Working	0.696	0.692	0.726	0.739	0.708
	(0.460)	(0.462)	(0.446)	(0.439)	(0.455)
Age Group of Child					
Preschool to	2,730	5,755	2,323	2,532	13,340
Elementary	(43.7%)	(45.2%)	(43.9%)	(42.6%)	(44.2%)
Middle to Junior High	1,797	3,483	1,589	1,850	8,719
	(28.7%)	(27.4%)	(30.0%)	(31.1%)	(28.9%)
High School	1,727	3,492	1,376	1,560	8,155
	(27.6%)	(27.4%)	(26.0%)	(26.3%)	(27.0%)
Trump Voter	0.406	0.397	0.393	0.390	0.398
	(0.491)	(0.489)	(0.488)	(0.488)	(0.489)

Appendix D: Weighted Tabulations

Tabulation of Race/Ethnicity (Weighted)

	Freq.	Percent	Cum.
1	14735.05	56.53	56.53
2	3931.481	15.08	71.61
3	6039.466	23.17	94.78
4	1359.999	5.22	100.00
Total	26066	100.00	

Tabulation of Race/Ethnicity

	Freq.	Percent	Cum.
1	20289	60.00	60.00
2	3424	10.13	70.13
3	7720	22.83	92.96
4	2381	7.04	100.00
Total	33814	100.00	

Tabulation of Gender (Weighted)

	Freq.	Percent	Cum.
0	15420.53	59.12	59.12
1	10661.47	40.88	100.00
Total	26082	100.00	

Tabulation of Gender

	Freq.	Percent	Cum.
0	21775	64.36	64.36
1	12057	35.64	100.00
Total	33832	100.00	

Tabulation of Married/Living with Partner (Weighted)

Marital Status	Freq.	Percent	Cum.
0	8918.359	34.19	34.19
Married	17163.64	65.81	100.00
Total	26082	100.00	

Tabulation of Married/Living with Partner

Marital Status	Freq.	Percent	Cum.
0	10654	31.49	31.49
Married	23178	68.51	100.00
Total	33832	100.00	

Tabulation of Education (Weighted)

Education	Freq.	Percent	Cum.
No College	9374.906	35.96	35.96
Some College	7852.34	30.12	66.08
College Graduate	8841.755	33.92	100.00
Total	26069	100.00	

Tabulation of Education

Education	Freq.	Percent	Cum.
No College	7454	22.04	22.04
Some College	11966	35.39	57.43
College Graduate	14395	42.57	100.00
Total	33815	100.00	

Tabulation of Age of Respondent (Weighted)

Age of Respondent	Freq.	Percent	Cum.
Under 30	1875.738	7.20	7.20
30 to 39	8236.617	31.60	38.80
40 to 49	9694.132	37.19	75.99
50 to 59	4211.214	16.16	92.14

60 and over	2048.3	7.86	100.00
Total	26066	100.00	

Tabulation of Age of Respondent

Age of Respondent	Freq.	Percent	Cum.
Under 30	1878	5.55	5.55
30 to 39	8077	23.89	29.44
40 to 49	13471	39.84	69.28
50 to 59	7230	21.38	90.66
60 and over	3159	9.34	100.00
Total	33815	100.00	

Tabulation of Medical Conditions (Weighted)

Medical Condition	Freq.	Percent	Cum.
No Immunocompromising Conditions	14120.01	54.15	54.15
Diagnosed Immunocompromising Medical Condition	11953.99	45.85	100.00
Total	26074	100.00	

Tabulation of Medical Conditions

Medical Condition	Freq.	Percent	Cum.
No Immunocompromising Conditions	17390	51.93	51.93
Diagnosed Immunocompromising Medical Condition	16095	48.07	100.00
Total	33485	100.00	

Tabulation of Urbanicity (Weighted)

Urbanicity	Freq.	Percent	Cum.
Rural	9517.866	36.49	36.49
Mixed	11105.25	42.58	79.07
Urban	5458.882	20.93	100.00
Total	26082	100.00	

Tabulation of Urbanicity

Urbanicity	Freq.	Percent	Cum.
Rural	12148	35.91	35.91
Mixed	12642	37.37	73.27
Urban	9042	26.73	100.00
Total	33832	100.00	

Tabulation of Age of Child (Weighted)

Age Group of Child	Freq.	Percent	Cum.
Preschool to Elementary	11039.6	49.13	49.13
Middle School to Junior High	6095.475	27.13	76.25
High School	5335.923	23.75	100.00
Total	22471	100.00	

Tabulation of Age of Child

Age Group of Child	Freq.	Percent	Cum.
Preschool to Elementary	13340	44.15	44.15
Middle School to Junior High	8719	28.86	73.01
High School	8155	26.99	100.00
Total	30214	100.00	

Tabulation of Income (Weighted)

Household Income	Freq.	Percent	Cum.
Low Income	11125.74	42.67	42.67
Middle Income	8254.3	31.66	74.33
High Income	6691.96	25.67	100.00
Total	26072	100.00	

Tabulation of Income

Household Income	Freq.	Percent	Cum.
Low Income	12337	36.48	36.48

Middle Income	10887	32.20	68.68
High Income	10591	31.32	100.00
Total	33815	100.00	

Tabulation of Number of Children (Weighted)

Number of Kids in the Household	Freq.	Percent	Cum.
1	12441.5	47.70	47.70
2	7115.814	27.28	74.98
3	4408.139	16.90	91.89
4	1269.41	4.87	96.75
5	523.6955	2.01	98.76
6	191.1034	0.73	99.49
7	110.0292	0.42	99.91
8	2.164442	0.01	99.92
10	20.14523	0.08	100.00
Total	26082	100.00	

Tabulation of Number of Children

Number of Kids in the Household	Freq.	Percent	Cum.
1	16713	49.40	49.40
2	10115	29.90	79.30
3	4866	14.38	93.68
4	1404	4.15	97.83
5	425	1.26	99.09
6	188	0.56	99.64
7	96	0.28	99.93
8	10	0.03	99.96
9	2	0.01	99.96
10	13	0.04	100.00
Total	33832	100.00	

Appendix E: Descriptive of Public Exits and Public Entrances

Table 1: Makeup of Public Exits by Phase

Race/Ethnicity	*Phase 1*	*Phase 2*	*Phase 3*	*Phase 4*	*Total Switchers*
White	86	45	32	19	182
Black	24	13	5	3	45
Hispanic/Latino	34	27	20	12	93
Asian/Other	9	4	5	4	22

Income	*Phase 1*	*Phase 2*	*Phase 3*	*Phase 4*	*Total Switchers*
Low	84	56	26	21	187
Medium	44	21	19	9	93
High	25	12	17	8	62

Education	*Phase 1*	*Phase 2*	*Phase 3*	*Phase 4*	*Total Switchers*
No College	56	26	16	11	109
Some College	55	39	25	17	136
4 Year College Degree	42	24	21	10	97

Table 2: Makeup of Public Exits by Type

Race/Ethnicity	*White*	*Black*	*Hisp/Lat*	*Asian/Other*	*Total Switchers*
Private	36	3	17	6	62
Homeschool	97	30	51	12	190
Other	49	12	25	4	90

Income	*Low*	*Middle*	*High*	*Total Switchers*
Private	19	21	22	62
Homeschool	112	50	28	190
Other	56	22	12	90

Education	*No College*	*Some College*	*4 -year College Degree*	*Total Switchers*
Private	7	22	33	62
Homeschool	72	79	39	190
Other	30	35	25	90

Table 3: Makeup of Public Entrances by Phase

Race/Ethnicity	*Phase 1*	*Phase 2*	*Phase 3*	*Phase 4*	*Total Switches*
White	75	65	55	29	224
Black	24	17	11	8	60
Hispanic/Latino	40	28	30	19	117
Asian/Other	14	8	5	9	36

Income	*Phase 1*	*Phase 2*	*Phase 3*	*Phase 4*	*Total Switches*
Low	94	66	42	29	231
Middle	38	26	32	13	109
High	21	26	27	23	97

Education	*Phase 1*	*Phase 2*	*Phase 3*	*Phase 4*	*Total Switches*
High School Degree	53	33	31	20	137
Some College	56	51	39	16	162
4 Year College Degree	44	34	31	29	138

Table 4: Makeup of Public-School Entrances by Type

Race/Ethnicity	*White*	*Black*	*Hisp/Lat*	*Asian/Other*	*Total Switchers*
Private	56	3	21	10	90
Homeschool	109	36	67	17	229
Other	59	21	29	9	118

Income	*Low*	*Middle*	*High*	*Total Switchers*
Private	28	21	41	90
Homeschool	132	60	37	229
Other	71	28	19	118

Education	*No College*	*Some College*	*4 -year College Degree*	*Total Switchers*
Private	16	30	44	90
Homeschool	85	90	54	229
Other	36	42	40	118

Table 5: General Tabulations

Tabulation of Public Exits by Type

	Phase				
	1	2	3	4	Total
Public	8078 (98.14%)	9665 (99.09%)	4048 (98.49%)	4607 (99.18%)	26398 (98.72%)
Private	22 (0.27%)	12 (0.12%)	15 (0.36%)	13 (0.28%)	62 (0.23%)
Homeschooled	95 (1.15%)	45 (0.46%)	36 (0.88%)	14 (0.30%)	190 (0.71%)
Other	36 (0.44%)	32 (0.33%)	11 (0.27%)	11 (0.24%)	90 (0.34%)
Total	8231	9754	4110	4645	26740

Tabulation of Public Entrance by Type

	Phase				
	1	2	3	4	Total
Pre-Return	1351 (89.83%)	1809 (93.88%)	577 (85.10%)	621 (90.52%)	4358 (90.89%)
Private to Public	22 (1.46%)	23 (1.19%)	23 (3.39%)	22 (3.2%)	90 (1.88%)
Homeschooled to Public	81 (5.39%)	62 (3.22%)	58 (8.55%)	28 (4.1%)	229 (4.78%)

Other to Public	50 (3.32%)	33 (1.71%)	20 (2.95%)	15 (2.19%)	118 (2.46%)
Total	1504	1927	678	686	4795

Tabulation of Public Exits

	Phase				
	1	2	3	4	Total
Public	8078 (98.14%)	9665 (99.09%)	4048 (98.49%)	4607 (99.18%)	26398 (98.72%)
Exit	153 (1.86%)	89 (0.91%)	62 (1.50%)	38 (0.82%)	342 (1.28%)
Total	8231	9754	4110	4645	26740

Tabulation of Public Entrance

	Phase				
	1	2	3	4	Total
Other than Public	1351 (89.83%)	1809 (93.88%)	577 (85.10%)	621 (90.52%)	4358 (90.89%)
Public Entrance	153 (10.17%)	118 (6.12%)	101 (14.90%)	65 (9.46%)	437 (9.11%)
Total	1504	1927	678	686	4795

Appendix F: Overall Multinomial Analysis by Phase

Phase 1:

	Phase 1			*Phase 1*			*Phase 1*			
	Set A			**Set B**			**Set C**			
VARIABLES	**P**	**H**	**O**	**P**	**H**	**O**	**P**	**H**	**O**	**Obs.**
UAS 264										6,213
UAS 270	0.000470	-0.00572	0.00257	0.000457	-0.00603	0.00258	0.000683	-0.00599	0.00259	6,213
	(0.00916)	(0.00834)	(0.00507)	(0.00912)	(0.00838)	(0.00512)	(0.00909)	(0.00838)	(0.00510)	6,213
UAS 272	0.000332	0.00529	0.000278	0.000847	0.00470	9.20e-05	0.00104	0.00469	0.000171	6,213
	(0.00915)	(0.00868)	(0.00492)	(0.00911)	(0.00869)	(0.00493)	(0.00908)	(0.00869)	(0.00493)	6,213
UAS 274	0.000396	-0.000700	-0.00187	0.00116	-0.00151	-0.00217	0.00144	-0.00150	-0.00208	6,213
	(0.00912)	(0.00846)	(0.00477)	(0.00907)	(0.00849)	(0.00476)	(0.00905)	(0.00848)	(0.00475)	6,213
Working	0.0263***	-0.0308***	-0.0103***	0.0275***	-0.0309***	-0.0103***	0.0275***	-0.0310***	-0.0104***	6,213
	(0.00840)	(0.00628)	(0.00393)	(0.00834)	(0.00629)	(0.00388)	(0.00835)	(0.00629)	(0.00387)	6,213
Gender	0.0149**	0.0247***	0.000705	0.00999	0.0240***	0.000563	0.00951	0.0235***	-0.000143	6,213
	(0.00685)	(0.00634)	(0.00394)	(0.00686)	(0.00646)	(0.00397)	(0.00683)	(0.00648)	(0.00399)	6,213
White										6,213
										6,213
Black	-0.0470***	0.00289	0.00800	-0.0475***	0.00209	0.00496	-0.0428***	0.00543	0.00770	6,213
	(0.00890)	(0.0117)	(0.00600)	(0.00907)	(0.0118)	(0.00549)	(0.00974)	(0.0125)	(0.00617)	6,213
Hispanic	-0.0169*	-0.0334***	0.00237	-0.0177*	-0.0326***	0.00265	-0.0133	-0.0308***	0.00457	6,213
	(0.00952)	(0.00772)	(0.00565)	(0.00951)	(0.00788)	(0.00592)	(0.00999)	(0.00814)	(0.00632)	6,213
Asian/Other	-0.0187*	-0.0468***	0.00667	-0.0245**	-0.0476***	0.00381	-0.0213*	-0.0463***	0.00560	6,213
	(0.0112)	(0.00906)	(0.00814)	(0.0107)	(0.00885)	(0.00739)	(0.0110)	(0.00891)	(0.00773)	6,213
Married/Living w/ Partner	-0.0106	0.0131*	-0.0146***	-0.0126	0.0119*	-0.0145***	-0.0128	0.0116*	-0.0148***	6,213
	(0.00803)	(0.00701)	(0.00415)	(0.00788)	(0.00710)	(0.00415)	(0.00789)	(0.00706)	(0.00414)	6,213
< High School										6,213
										6,213
High School Degree	0.0154*	0.000520	-0.0109**	0.0126	0.00303	-0.00949*	0.0128	0.00288	-0.00906*	6,213
	(0.00832)	(0.00823)	(0.00525)	(0.00795)	(0.00808)	(0.00518)	(0.00794)	(0.00806)	(0.00514)	6,213
College Degree	0.0479***	-0.0229***	-0.0146**	0.0539***	-0.0176**	-0.0130**	0.0540***	-0.0169*	-0.0123**	6,213

	(0.00934)	(0.00856)	(0.00565)	(0.00931)	(0.00870)	(0.00566)	(0.00931)	(0.00870)	(0.00563)	6,213
Age	0.0206***	0.000520	-0.000891	0.0190***	-0.000271	-0.00106	0.0186***	-0.000386	-0.00119	6,213
	(0.00405)	(0.00295)	(0.00183)	(0.00401)	(0.00301)	(0.00184)	(0.00398)	(0.00303)	(0.00186)	6,213
Medical Condition	-0.00243	0.000962	0.000303	0.000623	0.00252	0.00133	0.000836	0.00258	0.00135	6,213
	(0.00651)	(0.00611)	(0.00386)	(0.00658)	(0.00620)	(0.00389)	(0.00656)	(0.00621)	(0.00388)	6,213
Rural										6,213
										6,213
Mixed Environment	0.0545***	-0.0183**	0.00693	0.0555***	-0.0155*	0.00737	0.0565***	-0.0143*	0.00791	6,213
	(0.00966)	(0.00870)	(0.00565)	(0.00954)	(0.00866)	(0.00563)	(0.00960)	(0.00860)	(0.00551)	6,213
Urban Environment	0.0974***	-0.0167*	-0.00828	0.102***	-0.0136	-0.00785	0.105***	-0.0115	-0.00694	6,213
	(0.0128)	(0.0101)	(0.00536)	(0.0129)	(0.0102)	(0.00531)	(0.0133)	(0.0105)	(0.00540)	6,213
Missing Urbanicity	0.0814***	0.0111	-0.0157*	0.0805***	0.0105	-0.0164**	0.0917***	0.0154	-0.0136	6,213
	(0.0174)	(0.0102)	(0.00830)	(0.0172)	(0.0103)	(0.00828)	(0.0184)	(0.0114)	(0.00850)	6,213
Pre-K to K										6,213
										6,213
Middle School to Jr High	-0.0174**	-0.0353***	-0.00962**	-0.0159*	-0.0354***	-0.00985***	-0.0161**	-0.0355***	-0.00969**	6,213
	(0.00810)	(0.00672)	(0.00377)	(0.00813)	(0.00677)	(0.00380)	(0.00812)	(0.00676)	(0.00382)	6,213
High School	-0.0347***	-0.0306***	0.00364	-0.0341***	-0.0316***	0.00279	-0.0340***	-0.0318***	0.00280	6,213
	(0.00778)	(0.00725)	(0.00475)	(0.00780)	(0.00727)	(0.00474)	(0.00781)	(0.00726)	(0.00473)	6,213
Low Income										6,213
										6,213
Middle Income	0.00673	-0.00586	-0.00831**	0.00818	-0.00289	-0.00767*	0.00778	-0.00338	-0.00784**	6,213
	(0.00820)	(0.00779)	(0.00397)	(0.00818)	(0.00771)	(0.00398)	(0.00819)	(0.00771)	(0.00397)	6,213
High Income	0.0352***	-0.0377***	0.00233	0.0346***	-0.0347***	0.00353	0.0339***	-0.0351***	0.00326	6,213
	(0.00957)	(0.00803)	(0.00648)	(0.00937)	(0.00807)	(0.00677)	(0.00934)	(0.00807)	(0.00669)	6,213
Number of Kids	-0.0101**	0.0110***	-0.00255	-0.0103**	0.0111***	-0.00247	-0.0105**	0.0110***	-0.00250	6,213
	(0.00417)	(0.00245)	(0.00258)	(0.00409)	(0.00241)	(0.00252)	(0.00410)	(0.00241)	(0.00254)	6,213
Exposure to Remote	0.0103	-0.0105	0.00648*	0.0111	-0.0126*	0.00608	0.0117	-0.0130*	0.00612	6,213
	(0.00752)	(0.00724)	(0.00389)	(0.00748)	(0.00729)	(0.00392)	(0.00742)	(0.00725)	(0.00390)	6,213
Missing E Remote	0.0113	-0.0119	-0.243***	0.00512	-0.0124	-0.242***	0.00584	-0.0123	-0.241***	6,213
	(0.0393)	(0.0232)	(0.0232)	(0.0392)	(0.0233)	(0.0231)	(0.0388)	(0.0232)	(0.0231)	6,213
Perceived Risk				-0.00663*	0.000586	-0.00148	-0.00627*	0.000610	-0.00153	6,213
				(0.00382)	(0.00305)	(0.00173)	(0.00381)	(0.00304)	(0.00171)	6,213

Media Trust	-0.00204	0.00345	0.00532**	0.00240	0.00491	0.00660**	6,213
	(0.00407)	(0.00437)	(0.00270)	(0.00410)	(0.00437)	(0.00289)	6,213
Institutional Trust	-0.00948*	-0.0152***	-0.00528*	-0.00829*	-0.0147***	-0.00507*	6,213
	(0.00488)	(0.00479)	(0.00292)	(0.00491)	(0.00483)	(0.00290)	6,213
Trust in Fox News	0.0201***	0.0120***	0.00399**	0.0159***	0.0106***	0.00276	6,213
	(0.00358)	(0.00388)	(0.00173)	(0.00401)	(0.00392)	(0.00187)	6,213
Missing Percieved Risk	-0.783***	-0.718***	-0.212***	-0.779***	-0.717***	-0.211***	6,213
	(0.0504)	(0.0429)	(0.0227)	(0.0495)	(0.0432)	(0.0225)	6,213
Missing Media Trust	-0.798***	-0.681***	-0.215***	-0.788***	-0.677***	-0.212***	6,213
	(0.0706)	(0.0611)	(0.0268)	(0.0704)	(0.0611)	(0.0268)	6,213
Missing Trump Voter							6,213
							6,213
Trump Voter				0.0225***	0.00909	0.00707	6,213
				(0.00824)	(0.00806)	(0.00524)	6,213

Phase 2:

	Phase 2			*Phase 2*			*Phase 2*			
	Set A			**Set B**			**Set C**			
VARIABLES	**P**	**H**	**O**	**P**	**H**	**O**	**P**	**H**	**O**	**Obs.**
UAS 276										12,680
										12,680
UAS 278	0.00118	-0.0101	0.00309	0.000802	-0.00957	0.00283	0.00107	-0.00940	0.00281	12,680
	(0.00902)	(0.00771)	(0.00454)	(0.00914)	(0.00760)	(0.00452)	(0.00911)	(0.00759)	(0.00452)	12,680
UAS 280	0.00311	-0.00446	0.00634	0.00279	-0.00409	0.00612	0.00279	-0.00398	0.00611	12,680
	(0.00906)	(0.00790)	(0.00467)	(0.00919)	(0.00777)	(0.00467)	(0.00915)	(0.00776)	(0.00467)	12,680
UAS 282	0.00153	-0.0137*	0.00768	0.000317	-0.0124*	0.00750	0.000548	-0.0123	0.00749	12,680
	(0.00897)	(0.00760)	(0.00473)	(0.00905)	(0.00751)	(0.00470)	(0.00903)	(0.00750)	(0.00471)	12,680
UAS 340	-0.00141	-0.00312	0.00472	-0.00317	-0.00166	0.00482	-0.00309	-0.00145	0.00481	12,680
	(0.00893)	(0.00801)	(0.00453)	(0.00898)	(0.00794)	(0.00455)	(0.00895)	(0.00793)	(0.00455)	12,680
UAS 342	2.40e-05	-0.000719	0.00393	-0.00305	0.000892	0.00416	-0.00296	0.00111	0.00413	12,680
	(0.00904)	(0.00814)	(0.00460)	(0.00905)	(0.00808)	(0.00462)	(0.00903)	(0.00808)	(0.00462)	12,680
UAS 344	-0.00171	0.00222	0.00274	-0.00521	0.00409	0.00252	-0.00503	0.00419	0.00250	12,680
	(0.00901)	(0.00823)	(0.00453)	(0.00897)	(0.00819)	(0.00452)	(0.00895)	(0.00817)	(0.00452)	12,680
UAS 346	0.00122	0.0199**	0.0141**	-0.00237	0.0227**	0.0138**	-0.00243	0.0228**	0.0138**	12,680
Working	0.0259***	-0.0350***	-0.0197***	0.0267***	-0.0356***	-0.0198***	0.0268***	-0.0357***	-0.0198***	12,680
	(0.00603)	(0.00426)	(0.00284)	(0.00602)	(0.00427)	(0.00288)	(0.00601)	(0.00426)	(0.00287)	12,680
Gender	0.0125***	0.0226***	0.00741***	0.00932*	0.0222***	0.00701**	0.00889*	0.0217***	0.00706**	12,680
	(0.00483)	(0.00435)	(0.00271)	(0.00484)	(0.00434)	(0.00276)	(0.00481)	(0.00436)	(0.00279)	12,680
White										12,680
										12,680
Black	-0.0438***	-0.000523	-0.00761**	-0.0438***	-0.00831	-0.0104***	-0.0376***	-0.00479	-0.0106***	12,680
	(0.00641)	(0.00794)	(0.00380)	(0.00655)	(0.00758)	(0.00358)	(0.00716)	(0.00825)	(0.00382)	12,680
Hispanic	-0.00569	-0.0311***	-0.00986***	-0.00245	-0.0322***	-0.00996***	0.00376	-0.0303***	-0.0101***	12,680
	(0.00656)	(0.00492)	(0.00363)	(0.00667)	(0.00498)	(0.00364)	(0.00710)	(0.00520)	(0.00376)	12,680
Asian/Other	-0.00936	-0.0364***	-0.00300	-0.0142*	-0.0428***	-0.00537	-0.0101	-0.0413***	-0.00550	12,680
	(0.00808)	(0.00679)	(0.00496)	(0.00760)	(0.00629)	(0.00483)	(0.00785)	(0.00638)	(0.00495)	12,680
Married/Living w/ Partner	-0.00207	0.0137***	-0.00612**	-0.00401	0.0127***	-0.00699**	-0.00361	0.0125**	-0.00698**	12,680

	(0.00587)	(0.00489)	(0.00278)	(0.00581)	(0.00489)	(0.00276)	(0.00583)	(0.00488)	(0.00276)	12,680
< High School										12,680
										12,680
High School Degree	0.0130**	0.0207***	-0.00635*	0.0114**	0.0238***	-0.00433	0.0118**	0.0239***	-0.00438	12,680
	(0.00588)	(0.00545)	(0.00327)	(0.00581)	(0.00523)	(0.00327)	(0.00577)	(0.00520)	(0.00325)	12,680
College Degree	0.0524***	-0.00668	-0.00510	0.0548***	0.000967	-0.00144	0.0553***	0.00177	-0.00149	12,680
	(0.00674)	(0.00543)	(0.00383)	(0.00675)	(0.00565)	(0.00403)	(0.00671)	(0.00566)	(0.00401)	12,680
Age	0.0203***	-0.00377*	-0.00164	0.0195***	-0.00534***	-0.00190	0.0190***	-0.00540***	-0.00190	12,680
	(0.00286)	(0.00199)	(0.00118)	(0.00284)	(0.00203)	(0.00119)	(0.00282)	(0.00203)	(0.00120)	12,680
Medical Condition	-0.00527	-0.00181	-0.000232	-0.00138	0.00173	0.00106	-0.00114	0.00173	0.00106	12,680
	(0.00469)	(0.00420)	(0.00292)	(0.00473)	(0.00427)	(0.00296)	(0.00471)	(0.00427)	(0.00296)	12,680
Rural										12,680
										12,680
Mixed Environment	0.0510***	-0.0290***	0.0131***	0.0530***	-0.0254***	0.0138***	0.0546***	-0.0239***	0.0138***	12,680
	(0.00765)	(0.00622)	(0.00450)	(0.00763)	(0.00624)	(0.00452)	(0.00769)	(0.00616)	(0.00462)	12,680
Urban Environment	0.102***	-0.0315***	0.00731	0.106***	-0.0277***	0.00865*	0.111***	-0.0254***	0.00841*	12,680
	(0.00982)	(0.00693)	(0.00457)	(0.00982)	(0.00699)	(0.00463)	(0.0102)	(0.00712)	(0.00495)	12,680
Missing Urbanicity	0.0746***	-0.0143**	0.0132***	0.0781***	-0.0169**	0.0132***	0.0934***	-0.0117	0.0127**	12,680
	(0.0118)	(0.00645)	(0.00503)	(0.0118)	(0.00665)	(0.00503)	(0.0127)	(0.00742)	(0.00570)	12,680
Pre-K to K										12,680
										12,680
Middle School to Jr High	-0.00967*	-0.0386***	-0.0158***	-0.00971*	-0.0376***	-0.0152***	-0.00960	-0.0375***	-0.0151***	12,680
	(0.00587)	(0.00464)	(0.00278)	(0.00590)	(0.00465)	(0.00276)	(0.00591)	(0.00464)	(0.00275)	12,680
High School	-0.0338***	-0.0358***	-0.000919	-0.0349***	-0.0360***	-5.09e-05	-0.0345***	-0.0361***	-5.22e-05	12,680
	(0.00547)	(0.00495)	(0.00342)	(0.00547)	(0.00489)	(0.00344)	(0.00549)	(0.00488)	(0.00344)	12,680
Low Income										12,680
										12,680
Middle Income	0.00841	-9.16e-05	-0.0178***	0.00899	0.00784	-0.0147***	0.00784	0.00703	-0.0146***	12,680
	(0.00614)	(0.00555)	(0.00325)	(0.00614)	(0.00545)	(0.00314)	(0.00615)	(0.00541)	(0.00315)	12,680
High Income	0.0350***	-0.0321***	-0.0188***	0.0364***	-0.0235***	-0.0148***	0.0348***	-0.0240***	-0.0148***	12,680
	(0.00702)	(0.00538)	(0.00364)	(0.00695)	(0.00542)	(0.00369)	(0.00692)	(0.00542)	(0.00369)	12,680
Number of Kids	-0.00640**	0.00662***	-0.0113***	-0.00672**	0.00737***	-0.0102***	-0.00683**	0.00725***	-0.0102***	12,680
	(0.00294)	(0.00182)	(0.00255)	(0.00288)	(0.00175)	(0.00241)	(0.00289)	(0.00175)	(0.00240)	12,680

Exposure to Remote							12,680
							12,680
Missing E Remote							12,680
							12,680
Perceived Risk		-0.0133***	0.00506**	0.000828	-0.0128***	0.00518**	0.000813	12,680
		(0.00303)	(0.00214)	(0.00112)	(0.00302)	(0.00213)	(0.00112)	12,680
Media Trust		-0.000689	0.0104***	0.00606***	0.00507*	0.0118***	0.00594***	12,680
		(0.00284)	(0.00306)	(0.00177)	(0.00289)	(0.00309)	(0.00186)	12,680
Institutional Trust		-0.0100***	-0.0212***	-0.0101***	-0.00837**	-0.0208***	-0.0101***	12,680
		(0.00340)	(0.00337)	(0.00189)	(0.00341)	(0.00337)	(0.00188)	12,680
Trust in Fox News		0.0171***	0.0157***	0.00742***	0.0116***	0.0142***	0.00754***	12,680
		(0.00249)	(0.00257)	(0.00133)	(0.00276)	(0.00262)	(0.00149)	12,680
Missing Percieved Risk		0.0931	0.0536	-0.246***	0.0996	0.0554	-0.246***	12,680
		(0.0732)	(0.0522)	(0.0193)	(0.0717)	(0.0526)	(0.0193)	12,680
Missing Media Trust		-0.702***	-0.544***	-0.179***	-0.689***	-0.541***	-0.179***	12,680
		(0.0660)	(0.0513)	(0.0216)	(0.0657)	(0.0513)	(0.0216)	12,680
Missing Trump Voter								12,680
								12,680
Trump Voter					0.0308***	0.00958*	-0.000764	12,680
					(0.00614)	(0.00574)	(0.00407)	12,680

Phase 3:

	Phase 3			
	Set A			
VARIABLES	**P**	**H**	**O**	**Obs.**
UAS 350				5,208
				5,208
UAS 351	-0.00191	0.00887	-0.000745	5,208
	(0.00889)	(0.00658)	(0.00277)	5,208
UAS 461	-0.00662	0.000924	0.000858	5,208
	(0.00878)	(0.00625)	(0.00292)	5,208
Working	0.0131	-0.0289***	-0.00548**	5,208
	(0.00976)	(0.00582)	(0.00265)	5,208
Gender	-0.00634	0.0158***	0.00262	5,208
	(0.00783)	(0.00595)	(0.00283)	5,208
White				5,208
				5,208
Black	-0.0495***	0.00347	0.00684	5,208
	(0.00888)	(0.0111)	(0.00508)	5,208
Hispanic	0.00793	-0.0237***	-0.00182	5,208
	(0.0108)	(0.00616)	(0.00289)	5,208
Asian/Other	0.00537	-0.00808	-0.00498	5,208
	(0.0140)	(0.0112)	(0.00312)	5,208
Married/Living w/ Partner	-0.0114	0.0201***	0.00403	5,208
	(0.00875)	(0.00730)	(0.00291)	5,208
< High School				5,208
				5,208
High School Degree	0.0185**	0.0156**	-0.00285	5,208
	(0.00920)	(0.00714)	(0.00322)	5,208
College Degree	0.0447***	-0.000412	-0.00305	5,208
	(0.0101)	(0.00747)	(0.00364)	5,208
Age	0.0215***	-0.00267	-0.00101	5,208
	(0.00421)	(0.00280)	(0.00129)	5,208

Medical Condition	0.00668	0.00605	0.000719	5,208
	(0.00740)	(0.00552)	(0.00237)	5,208
Rural				5,208
				5,208
Mixed Environment	0.0493***	0.00789	0.00287	5,208
	(0.0148)	(0.00816)	(0.00491)	5,208
Urban Environment	0.0960***	0.00471	-0.00176	5,208
	(0.0190)	(0.00914)	(0.00364)	5,208
Missing Urbanicity	0.0613***	0.00669	0.00681*	5,208
	(0.0176)	(0.00868)	(0.00404)	5,208
Pre-K to K				5,208
				5,208
Middle School to Jr High	-0.0441***	0.00316	-0.000552	5,208
	(0.00887)	(0.00664)	(0.00273)	5,208
High School	-0.0532***	-0.0115*	0.00348	5,208
	(0.00902)	(0.00638)	(0.00341)	5,208
Low Income				5,208
				5,208
Middle Income	0.00171	0.000356	-0.00543	5,208
	(0.00932)	(0.00817)	(0.00437)	5,208
High Income	0.0353***	-0.0252***	-0.0101***	5,208
	(0.0109)	(0.00785)	(0.00362)	5,208
Number of Kids	-0.0124***	0.00837***	-0.00419	5,208
	(0.00416)	(0.00207)	(0.00269)	5,208
Exposure to Remote				5,208

Phase 4:

	Phase 4			*Phase 4*			*Phase 4*			
	Set A			**Set B**			**Set C**			
VARIABLES	**P**	**H**	**O**	**P**	**H**	**O**	**P**	**H**	**O**	**Obs.**
UAS 475										4,339
										4,339
UAS 502	0.00503	-0.00548	-0.00446	0.00489	-0.00265	-0.00510	0.00496	-0.00261	-0.00502	4,339
	(0.00881)	(0.00579)	(0.00314)	(0.00944)	(0.00626)	(0.00312)	(0.00943)	(0.00627)	(0.00308)	4,339
UAS 559	0.0101	0.00178	-0.00481	0.0119	0.00407	-0.00562*	0.0120	0.00405	-0.00548*	4,339
	(0.00865)	(0.00604)	(0.00305)	(0.00968)	(0.00664)	(0.00315)	(0.00967)	(0.00665)	(0.00313)	4,339
Working	0.0104	-0.0150***	-0.00537**	0.0169	-0.0144**	-0.00523**	0.0169	-0.0145**	-0.00478**	4,339
	(0.00962)	(0.00535)	(0.00229)	(0.0110)	(0.00596)	(0.00240)	(0.0110)	(0.00594)	(0.00242)	4,339
Gender	-0.00958	0.0170***	0.00238	0.0106	0.0154**	0.00119	0.0109	0.0153**	0.00152	4,339
	(0.00794)	(0.00541)	(0.00233)	(0.00854)	(0.00617)	(0.00250)	(0.00852)	(0.00620)	(0.00254)	4,339
White										4,339
										4,339
Black	-0.0448***	-0.0124*	-0.00214	-0.0500***	-0.0150*	-0.00245	-0.0455***	-0.0126	-0.00484	4,339
	(0.00947)	(0.00738)	(0.00332)	(0.0106)	(0.00887)	(0.00381)	(0.0112)	(0.00987)	(0.00369)	4,339
Hispanic	-0.00345	-0.0172***	-0.00141	0.00136	-0.0223***	-0.00261	0.00632	-0.0209***	-0.00434	4,339
	(0.0104)	(0.00595)	(0.00304)	(0.0121)	(0.00658)	(0.00360)	(0.0126)	(0.00702)	(0.00357)	4,339
Asian/Other	0.0103	-0.0107	-0.00844***	0.000284	-0.0167*	-0.00801***	0.00333	-0.0158	-0.00939***	4,339
	(0.0138)	(0.00891)	(0.00193)	(0.0149)	(0.00977)	(0.00241)	(0.0151)	(0.00970)	(0.00274)	4,339
Married/Living w/ Partner	-0.0102	0.00818	-0.000399	-0.0259***	0.0162**	0.00125	-0.0255***	0.0160**	0.00129	4,339
	(0.00855)	(0.00566)	(0.00244)	(0.00952)	(0.00695)	(0.00248)	(0.00959)	(0.00696)	(0.00248)	4,339
< High School										4,339
										4,339
High School Degree	0.0146	0.0146**	-0.00445	-0.000211	0.0152**	-0.00529	-0.000238	0.0148**	-0.00518	4,339
	(0.00890)	(0.00577)	(0.00289)	(0.00946)	(0.00624)	(0.00374)	(0.00936)	(0.00631)	(0.00383)	4,339
College Degree	0.0528***	0.00874	-0.00465	0.0604***	0.0115	-0.00898**	0.0613***	0.0117	-0.00912**	4,339
	(0.0102)	(0.00643)	(0.00347)	(0.0121)	(0.00743)	(0.00396)	(0.0120)	(0.00744)	(0.00414)	4,339
Age	0.0203***	-0.00387	-0.00217*	0.0189***	-0.00327	-0.00200	0.0187***	-0.00331	-0.00188	4,339
	(0.00412)	(0.00270)	(0.00123)	(0.00464)	(0.00337)	(0.00142)	(0.00461)	(0.00337)	(0.00140)	4,339

Medical Condition	0.00660	5.22e-05	0.00128	0.00533	-0.00295	0.000894	0.00493	-0.00327	0.000821	4,339
	(0.00733)	(0.00478)	(0.00230)	(0.00838)	(0.00559)	(0.00243)	(0.00835)	(0.00556)	(0.00241)	4,339
Rural										4,339
										4,339
Mixed Enviroment	0.0530***	-0.0219***	0.000621	0.0465***	-0.0196***	0.000868	0.0484***	-0.0186**	-0.000214	4,339
	(0.0177)	(0.00701)	(0.00439)	(0.0132)	(0.00756)	(0.00388)	(0.0132)	(0.00747)	(0.00408)	4,339
Urban Enviroment	0.108***	-0.0198**	-0.000373	0.0970***	-0.0125	4.34e-05	0.102***	-0.0108	-0.00148	4,339
	(0.0229)	(0.00799)	(0.00418)	(0.0169)	(0.00907)	(0.00408)	(0.0175)	(0.00921)	(0.00422)	4,339
Missing Urbanicity	0.0692***	-0.0168**	0.00311	0.0673***	-0.0171**	0.00118	0.0801***	-0.0123	-0.00211	4,339
	(0.0176)	(0.00661)	(0.00409)	(0.0187)	(0.00862)	(0.00455)	(0.0199)	(0.00968)	(0.00470)	4,339
Pre-K to K										4,339
										4,339
Middle School to Jr High	-0.0267***	-0.00465	0.00164	-0.0450***	-0.00562	0.000357	-0.0450***	-0.00574	0.000452	4,339
	(0.00880)	(0.00553)	(0.00239)	(0.00994)	(0.00625)	(0.00255)	(0.00991)	(0.00627)	(0.00254)	4,339
High School	-0.0424***	-0.00140	0.00899**	-0.0604***	-0.00282	0.00680*	-0.0606***	-0.00309	0.00698*	4,339
	(0.00870)	(0.00614)	(0.00358)	(0.00989)	(0.00673)	(0.00381)	(0.00989)	(0.00674)	(0.00392)	4,339
Low Income										4,339
										4,339
Middle Income	-0.00736	-0.00576	-0.00434	-0.00612	-0.0120	-0.00304	-0.00775	-0.0130	-0.00240	4,339
	(0.00973)	(0.00744)	(0.00319)	(0.0113)	(0.00897)	(0.00325)	(0.0114)	(0.00894)	(0.00347)	4,339
High Income	0.0222**	-0.0285***	-0.00927***	0.0168	-0.0338***	-0.00544*	0.0145	-0.0347***	-0.00486	4,339
	(0.0112)	(0.00736)	(0.00280)	(0.0126)	(0.00864)	(0.00307)	(0.0127)	(0.00863)	(0.00330)	4,339
Number of Kids	-0.00374	0.00858***	-0.00195	-0.00382	0.0100***	-0.00266	-0.00400	0.00988***	-0.00274	4,339
	(0.00360)	(0.00183)	(0.00173)	(0.00402)	(0.00190)	(0.00181)	(0.00404)	(0.00190)	(0.00187)	4,339
Exposure to Remote										4,339
										4,339
Missing E Remote										4,339
										4,339
Perceived Risk										4,339
										4,339
Media Trust				-0.00998**	-0.00376	0.00265	-0.00554	-0.00257	0.00146	4,339
				(0.00498)	(0.00421)	(0.00194)	(0.00514)	(0.00427)	(0.00191)	4,339
Institutional Trust				7.95e-06	-0.00921**	-0.00245	0.000736	-0.00880**	-0.00260	4,339

	(0.00571)	(0.00407)	(0.00222)	(0.00563)	(0.00411)	(0.00228)	4,339
Trust in Fox News	0.0231***	0.00434	-0.00159	0.0187***	0.00306	-0.000466	4,339
	(0.00454)	(0.00341)	(0.00132)	(0.00502)	(0.00355)	(0.00139)	4,339
Missing Perceived Risk							4,339
							4,339
Missing Media Trust	-0.881***	-0.359***	-0.0637***	-0.924***	-0.379***	-0.0693***	4,339
	(0.0652)	(0.0380)	(0.0150)	(0.0671)	(0.0394)	(0.0158)	4,339
Missing Trump Voter							4,339
							4,339
Trump Voter				0.0242**	0.00777	-0.00924*	4,339
				(0.0102)	(0.00727)	(0.00509)	4,339

Appendix G: Public Exit by Phase

		Set A		Set B		Set C	
VARIABLES	Overall	Phase 1	Phase 2	Phase 1	Phase 2	Phase 1	Phase 2
UAS 270	-0.0239***	-0.0244***		-0.0244***		-0.0244***	
	(0.00741)	(0.00755)		(0.00760)		(0.00760)	
UAS 272	-0.0318***	-0.0324***		-0.0325***		-0.0325***	
	(0.00697)	(0.00707)		(0.00702)		(0.00703)	
UAS 274	-0.0361***	-0.0369***		-0.0371***		-0.0371***	
	(0.00682)	(0.00693)		(0.00690)		(0.00691)	
UAS 276	-0.0369***						
	(0.00678)						
UAS 278	-0.0388***		-0.00271		-0.00205		-0.00210
	(0.00664)		(0.00412)		(0.00401)		(0.00400)
UAS 280	-0.0417***		-0.00578		-0.00494		-0.00495
	(0.00646)		(0.00386)		(0.00374)		(0.00374)
UAS 282	-0.0440***		-0.00800**		-0.00711**		-0.00713**
	(0.00634)		(0.00363)		(0.00353)		(0.00353)
UAS 340	-0.0394***		-0.00351		-0.00262		-0.00264
	(0.00661)		(0.00405)		(0.00398)		(0.00397)
UAS 342	-0.0382***		-0.00207		-0.00113		-0.00112
	(0.00671)		(0.00425)		(0.00418)		(0.00418)
UAS 344	-0.0404***		-0.00417		-0.00332		-0.00332
	(0.00659)		(0.00410)		(0.00400)		(0.00400)
UAS 346	-0.0378***		-0.00182		-0.000502		-0.000495
	(0.00706)		(0.00476)		(0.00476)		(0.00476)
UAS 350	-0.0269***						
	(0.00739)						
UAS 351	-0.0343***						
	(0.00702)						
UAS 461	-0.0377***						
	(0.00680)						
UAS 475	-0.0467***						
	(0.00621)						
UAS 502	-0.0382***						
	(0.00684)						
UAS 559	-0.0377***						
	(0.00695)						
Working	-0.00336*	-0.00629	-0.00125	-0.00555	-0.00189	-0.00559	-0.00179
	(0.00175)	(0.00509)	(0.00200)	(0.00513)	(0.00203)	(0.00516)	(0.00204)
Gender	-0.000487	-0.00185	-0.000542	-0.00128	5.41e-05	-0.00134	0.000231
	(0.00184)	(0.00520)	(0.00214)	(0.00527)	(0.00215)	(0.00532)	(0.00218)
Black	-0.00220	0.00264	0.000599	0.00175	-0.00149	0.00211	-0.00271
	(0.00258)	(0.00810)	(0.00310)	(0.00810)	(0.00293)	(0.00820)	(0.00291)

Hispanic	-0.000392	-0.00157	-0.00184	-0.00245	-0.00179	-0.00219	-0.00285
	(0.00248)	(0.00726)	(0.00253)	(0.00727)	(0.00268)	(0.00762)	(0.00281)
Asian/Other	-0.00586**	-0.0113	-0.00478	-0.0125*	-0.00731***	-0.0123	-0.00800***
	(0.00272)	(0.00763)	(0.00311)	(0.00746)	(0.00255)	(0.00762)	(0.00258)
Married/Living w/ Partner	0.000771	-0.00109	0.00192	-0.00115	0.00136	-0.00117	0.00127
	(0.00183)	(0.00516)	(0.00207)	(0.00519)	(0.00207)	(0.00518)	(0.00206)
< High School							
High School Degree	-0.000395	-0.00725	0.00146	-0.00597	0.00260	-0.00593	0.00247
	(0.00217)	(0.00650)	(0.00256)	(0.00645)	(0.00247)	(0.00647)	(0.00246)
College Degree	-0.00326	-0.0120*	-0.00134	-0.00973	0.000965	-0.00964	0.000738
	(0.00238)	(0.00722)	(0.00262)	(0.00729)	(0.00275)	(0.00731)	(0.00277)
Age	0.000358	0.00338	7.63e-06	0.00231	-0.000549	0.00229	-0.000528
	(0.000903)	(0.00255)	(0.00107)	(0.00261)	(0.00108)	(0.00261)	(0.00108)
Medical Condition	0.00177	0.00410	-0.00189	0.00522	-0.000757	0.00524	-0.000797
	(0.00169)	(0.00477)	(0.00209)	(0.00485)	(0.00214)	(0.00485)	(0.00215)
Rural							
Mixed Environment	5.35e-05	-0.00497	-0.000565	-0.00388	0.000447	-0.00379	9.17e-05
	(0.00239)	(0.00682)	(0.00285)	(0.00679)	(0.00279)	(0.00677)	(0.00296)
Urban Environment	0.00192	-0.00676	0.00487	-0.00553	0.00658*	-0.00530	0.00529
	(0.00282)	(0.00772)	(0.00372)	(0.00763)	(0.00393)	(0.00776)	(0.00404)
Missing Urbanicity	0.00238	0.00266	0.00375	0.00266	0.00288	0.00317	0.000930
	(0.00299)	(0.00827)	(0.00346)	(0.00840)	(0.00359)	(0.00874)	(0.00399)
Pre-K to K							
Middle School to Jr High	-0.000336	-0.00854	0.000887	-0.00836	0.00158	-0.00834	0.00152
	(0.00192)	(0.00541)	(0.00231)	(0.00543)	(0.00236)	(0.00544)	(0.00237)
High School	-0.000898	-0.00738	0.00147	-0.00749	0.00180	-0.00749	0.00180
	(0.00204)	(0.00578)	(0.00247)	(0.00587)	(0.00244)	(0.00587)	(0.00244)
Middle Income	-0.00650***	-0.0109*	-0.00905***	-0.00933	-0.00691***	-0.00936	-0.00663**
	(0.00224)	(0.00617)	(0.00270)	(0.00615)	(0.00258)	(0.00618)	(0.00261)
High Income	-0.00884***	-0.0155**	-0.0116***	-0.0139**	-0.00952***	-0.0139**	-0.00928***
	(0.00243)	(0.00689)	(0.00267)	(0.00704)	(0.00262)	(0.00704)	(0.00264)
Number of Kids	-0.000436	0.00137	-0.00138	0.00135	-0.00147	0.00135	-0.00143
	(0.000906)	(0.00253)	(0.00141)	(0.00250)	(0.00137)	(0.00250)	(0.00136)
Trust in Fox News	0.00450***			0.00913***	0.00520***	0.00897***	0.00570***
	(0.00104)			(0.00287)	(0.00117)	(0.00294)	(0.00128)
Trump Voter	-0.00158					0.00105	-0.00387
	(0.00234)					(0.00633)	(0.00307)
Media Trust	0.00115			0.000963	0.00295**	0.00112	0.00243*
	(0.00116)			(0.00319)	(0.00142)	(0.00329)	(0.00144)
Institutional Trust	-0.00393***			-0.00231	-0.00686***	-0.00226	-0.00688***
	(0.00121)			(0.00362)	(0.00154)	(0.00361)	(0.00154)
Exposure to Remote		-0.00266		-0.00295		-0.00295	
		(0.00583)		(0.00575)		(0.00575)	

Missing E Remote		-		-		-	
Perceived Risk				0.000489	-0.000375	0.000498	-0.000427
				(0.00223)	(0.00104)	(0.00223)	(0.00103)
Observations	21,579	4,974	9,705	4,956	9,698	4,956	9,698

Appendix H: Public Return By Phase

		Set A				*Set B*				*Set C*			
VARIABLES	Overall	Phase 1	Phase 2	Phase 3	Phase 4	Phase 1	Phase 2	Phase 3	Phase 4	Phase 1	Phase 2	Phase 3	Phase 4
UAS 270	0.0556*	0.0547*				0.0584**				0.0573**			
	(0.0290)	(0.0295)				(0.0294)				(0.0292)			
UAS 272	-0.0331	-0.0323				-0.0283				-0.0277			
	(0.0261)	(0.0280)				(0.0278)				(0.0277)			
UAS 274	-0.0158	-0.0178				-0.0141				-0.0124			
	(0.0271)	(0.0285)				(0.0283)				(0.0283)			
UAS 276	-0.0406												
	(0.0263)												
UAS 278	-0.0418		0.000417				-0.000853				-0.00211		
	(0.0263)		(0.0224)				(0.0220)				(0.0219)		
UAS 280	-0.0489*		-0.00824				-0.00833				-0.00908		
	(0.0259)		(0.0219)				(0.0216)				(0.0216)		
UAS 282	-0.0549**		-0.00812				-0.0103				-0.0108		
	(0.0254)		(0.0220)				(0.0218)				(0.0218)		
UAS 340	-0.0705***		-0.0275				-0.0265				-0.0283		
	(0.0245)		(0.0204)				(0.0204)				(0.0201)		
UAS 342	-0.0496*		-0.00577				-0.00417				-0.00651		
	(0.0260)		(0.0224)				(0.0225)				(0.0221)		
UAS 344	-0.0415		0.00122				0.00474				0.00598		
	(0.0266)		(0.0228)				(0.0234)				(0.0235)		
UAS 346	-0.0681***		-0.0255				-0.0255				-0.0245		
	(0.0256)		(0.0215)				(0.0216)				(0.0220)		
UAS 350	0.0924***												
	(0.0333)												
UAS 351	0.0297			-0.0541*				-0.0464*				-0.0593*	
	(0.0317)			(0.0329)				(0.0259)				(0.0351)	
UAS 461	0.00666			-0.0784**				0.605***				-0.0837**	

	(0.0320)			(0.0335)				(0.0204)				(0.0362)	
UAS 475	-0.0636**												
	(0.0272)												
UAS 502	0.0655*				0.133***				0.118***				0.119***
	(0.0352)				(0.0277)				(0.0264)				(0.0262)
UAS 559	-0.0557**				0.0149				0.0297				0.0302
	(0.0284)				(0.0208)				(0.0206)				(0.0208)
Working	-0.00267	0.0283	0.0109	-0.0219	-0.0115	0.0298	0.0109	-0.0221	-0.0445	0.0271	0.00912	-0.0334	-0.0455
	(0.0104)	(0.0231)	(0.0133)	(0.0308)	(0.0265)	(0.0232)	(0.0133)	(0.0349)	(0.0286)	(0.0234)	(0.0132)	(0.0349)	(0.0287)
Gender	-0.0315***	-0.0331	-0.0399***	-0.0211	0.00627	-0.0256	-0.0421***	-0.0358	0.000200	-0.0258	-0.0424***	-0.0364	0.00157
	(0.0104)	(0.0224)	(0.0132)	(0.0294)	(0.0230)	(0.0228)	(0.0135)	(0.0325)	(0.0257)	(0.0229)	(0.0138)	(0.0325)	(0.0264)
Black	0.0321*	0.0780*	0.0201	0.0810	0.0805	0.0754*	0.00949	0.0492	0.239***	0.0550	-0.00244	0.0305	0.250***
	(0.0189)	(0.0433)	(0.0227)	(0.0630)	(0.0550)	(0.0451)	(0.0224)	(0.0673)	(0.0829)	(0.0420)	(0.0206)	(0.0656)	(0.0861)
Hispanic/Latino	0.0273*	0.0674*	0.00804	0.0757	0.0827**	0.0642*	0.00200	0.0360	0.0618*	0.0569*	-0.00345	0.0365	0.0665*
	(0.0143)	(0.0352)	(0.0178)	(0.0460)	(0.0340)	(0.0355)	(0.0181)	(0.0487)	(0.0351)	(0.0345)	(0.0173)	(0.0478)	(0.0378)
Asian/Other	-0.000551	0.0535	-0.0106	-0.0118	0.0576	0.0577	-0.0194	-0.0332	0.0747	0.0453	-0.0258	-0.0442	0.0787
	(0.0178)	(0.0504)	(0.0194)	(0.0557)	(0.0470)	(0.0520)	(0.0185)	(0.0559)	(0.0545)	(0.0508)	(0.0175)	(0.0535)	(0.0574)
Married/Living w/ Partner	0.00862	0.0488**	-0.00852	-0.0406	-0.0148	0.0460*	-0.00314	-0.0515	0.0545	0.0533**	0.000879	-0.0549	0.0517
	(0.0115)	(0.0241)	(0.0134)	(0.0338)	(0.0302)	(0.0246)	(0.0138)	(0.0373)	(0.0382)	(0.0250)	(0.0141)	(0.0371)	(0.0398)
< High School													
High School Degree	-0.0450***	-0.0283	-0.00912	-0.115**	-0.111**	-0.0292	-0.0101	-0.100**	-0.147***	-0.0317	-0.0206	-0.109**	-0.142***
	(0.0145)	(0.0296)	(0.0172)	(0.0498)	(0.0440)	(0.0302)	(0.0168)	(0.0509)	(0.0504)	(0.0301)	(0.0181)	(0.0503)	(0.0484)
College Degree	-0.0684***	-0.0554*	-0.0365**	-0.159***	-0.101**	-0.0658**	-0.0287	-0.136**	-0.132**	-0.0683**	-0.0391**	-0.147***	-0.127**
	(0.0158)	(0.0307)	(0.0183)	(0.0529)	(0.0460)	(0.0320)	(0.0187)	(0.0574)	(0.0521)	(0.0320)	(0.0191)	(0.0551)	(0.0511)
Age	-0.00383	0.000230	-0.00544	-0.00766	0.00265	0.00167	-0.00635	-0.0139	-0.0137	0.00260	-0.00608	-0.0118	-0.0150
	(0.00486)	(0.0102)	(0.00566)	(0.0153)	(0.0129)	(0.0103)	(0.00586)	(0.0169)	(0.0151)	(0.0103)	(0.00578)	(0.0166)	(0.0147)
Medical Condition	0.00261	-0.0359	0.0219*	-0.00904	-0.0127	-0.0354	0.0224*	0.00469	0.00227	-0.0336	0.0246*	0.00369	0.00284
	(0.0101)	(0.0223)	(0.0123)	(0.0297)	(0.0242)	(0.0226)	(0.0129)	(0.0334)	(0.0276)	(0.0225)	(0.0128)	(0.0333)	(0.0272)
Rural													

Mixed Environment	-0.00175	-0.0456	-0.00167	0.0400	0.118*	-0.0468	-0.000544	0.0466	0.113**	-0.0516	-0.00589	0.0333	0.116***
	(0.0161)	(0.0316)	(0.0181)	(0.0596)	(0.0665)	(0.0319)	(0.0183)	(0.0616)	(0.0445)	(0.0321)	(0.0191)	(0.0624)	(0.0444)
Urban Environment	-0.00956	-0.0138	0.0102	-0.0198	0.0314	-0.0184	0.0123	-0.00609	0.0398	-0.0252	0.00432	-0.0273	0.0423
	(0.0177)	(0.0383)	(0.0215)	(0.0597)	(0.0541)	(0.0380)	(0.0218)	(0.0641)	(0.0440)	(0.0383)	(0.0221)	(0.0646)	(0.0430)
Missing Urbanicity	0.000181	0.00687	0.0176	0.0208	0.0711	0.00927	0.0132	0.0809	0.0820	-0.0182	-0.00910	0.0178	0.0940
Pre-K to K													
	(0.0189)	(0.0356)	(0.0213)	(0.0588)	(0.0562)	(0.0353)	(0.0218)	(0.0653)	(0.0578)	(0.0374)	(0.0234)	(0.0687)	(0.0614)
Middle School to Jr High	0.0499***	0.0570**	0.0273*	0.0230	0.0505*	0.0553**	0.0263*	0.0271	0.116***	0.0564**	0.0297*	0.0365	0.117***
	(0.0125)	(0.0269)	(0.0155)	(0.0349)	(0.0288)	(0.0270)	(0.0154)	(0.0382)	(0.0373)	(0.0272)	(0.0157)	(0.0390)	(0.0374)
High School	0.0143	0.0292	0.0137	-0.0235	0.0240	0.0299	0.0136	-0.0302	0.0452	0.0317	0.0134	-0.0327	0.0474
	(0.0120)	(0.0270)	(0.0152)	(0.0366)	(0.0301)	(0.0273)	(0.0152)	(0.0396)	(0.0329)	(0.0274)	(0.0153)	(0.0390)	(0.0337)
Low Income													
Middle Income	-0.0234*	-0.0705**	-0.0313**	-0.00848	-0.0293	-0.0737**	-0.0241	0.0102	-0.0323	-0.0759**	-0.0181	0.0312	-0.0350
	(0.0135)	(0.0319)	(0.0157)	(0.0389)	(0.0344)	(0.0330)	(0.0159)	(0.0423)	(0.0399)	(0.0328)	(0.0163)	(0.0420)	(0.0407)
High Income	-0.0338**	-0.125***	-0.0200	-0.0129	-0.0191	-0.132***	-0.0110	0.0136	-0.0396	-0.131***	-0.00299	0.0314	-0.0425
	(0.0152)	(0.0306)	(0.0204)	(0.0465)	(0.0369)	(0.0314)	(0.0211)	(0.0525)	(0.0416)	(0.0317)	(0.0214)	(0.0511)	(0.0423)
Number of Kids	-0.00582	-0.00957	-0.00310	0.00436	-0.0139	-0.0120	-0.000829	0.00515	-0.0285*	-0.0118	-0.00131	0.00491	-0.0277*
	(0.00464)	(0.00846)	(0.00642)	(0.0117)	(0.0125)	(0.00913)	(0.00633)	(0.0130)	(0.0149)	(0.00909)	(0.00654)	(0.0132)	(0.0152)
Media Trust	0.00109					0.00134	0.0132	0.0247	0.0319*	-0.00996	0.00379	0.00375	0.0379*
	(0.00791)					(0.0160)	(0.00871)	(0.0255)	(0.0184)	(0.0177)	(0.00930)	(0.0274)	(0.0224)
Institutional Trust	-0.0143*					0.0141	-0.0238**	-0.0210	-0.0333*	0.0114	-0.0265***	-0.0297	-0.0330*
	(0.00803)					(0.0170)	(0.00963)	(0.0280)	(0.0202)	(0.0169)	(0.0102)	(0.0285)	(0.0199)
Trust in Fox News	0.00720					-0.0180	0.00899	0.00947	-0.0302**	-0.00689	0.0178**	0.0322	-0.0331**
	(0.00636)					(0.0127)	(0.00684)	(0.0195)	(0.0147)	(0.0145)	(0.00814)	(0.0211)	(0.0161)
Trump Voter	-0.0539***									-0.0571*	-0.0481***	-0.115**	0.0233
	(0.0148)									(0.0331)	(0.0181)	(0.0504)	(0.0444)
Exposure to Remote		-0.0260				-0.0227				-0.0232			
		(0.0247)				(0.0246)				(0.0245)			
Perceived Risk						-0.00814	0.00745	0.0181		-0.00755	0.00650		

						(0.00984)	(0.00508)	(0.0190)		(0.00996)	(0.00506)		
Missing Perceived Risk								-1.512***					
								(0.192)					
Observations	4,006	1,035	1,906	673	655	1,030	1,900	590	475	1,030	1,900	590	475

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Appendix I: Public Exit by Type

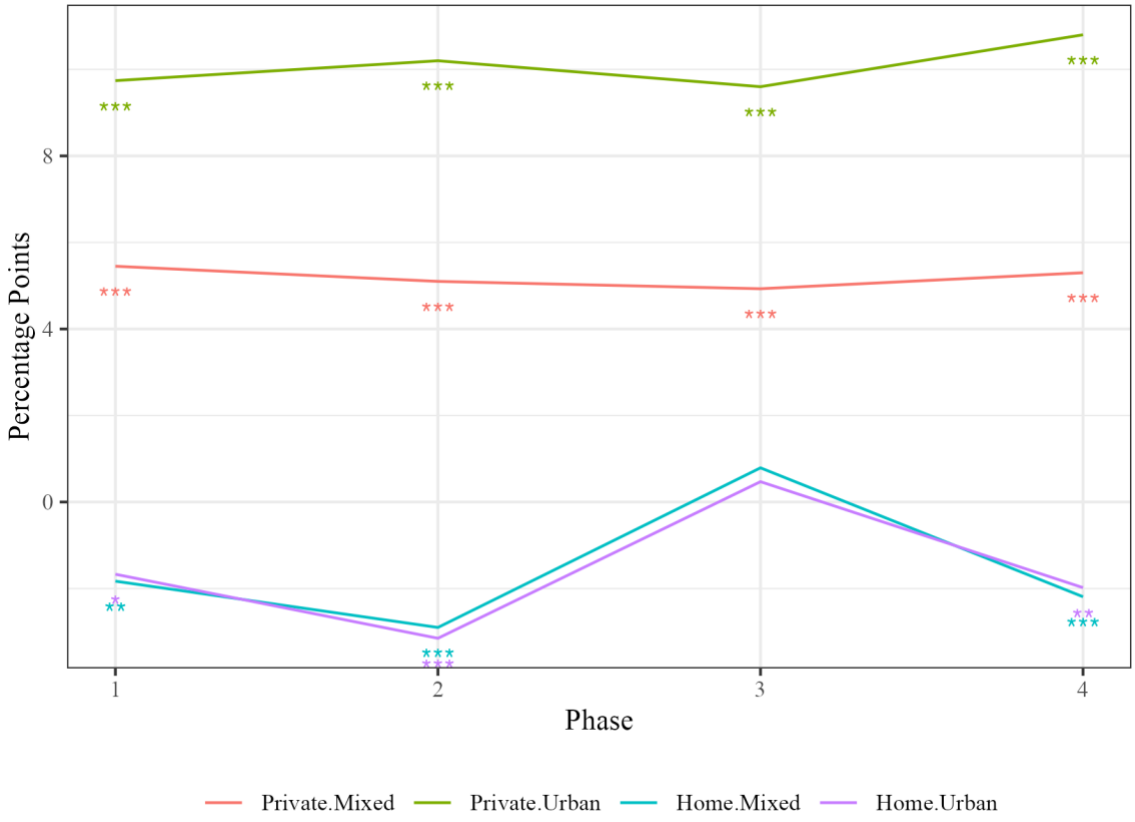
	Private		*Home*		*Other*		*Obs.*
UAS 264							21,584
UAS 270	-0.00360	(0.00309)	-0.0165***	(0.00574)	-0.00377	(0.00367)	21,584
UAS 272	-0.00515*	(0.00289)	-0.0184***	(0.00558)	-0.00820***	(0.00316)	21,584
UAS 274	-0.00601**	(0.00278)	-0.0234***	(0.00529)	-0.00655*	(0.00340)	21,584
UAS 276	-0.00439	(0.00298)	-0.0250***	(0.00521)	-0.00735**	(0.00328)	21,584
UAS 278	-0.00601**	(0.00277)	-0.0260***	(0.00511)	-0.00683**	(0.00332)	21,584
UAS 280	-0.00760***	(0.00255)	-0.0267***	(0.00505)	-0.00740**	(0.00322)	21,584
UAS 282	-0.00608**	(0.00277)	-0.0297***	(0.00481)	-0.00813**	(0.00319)	21,584
UAS 340	-0.00760***	(0.00255)	-0.0236***	(0.00528)	-0.00814**	(0.00318)	21,584
UAS 342	-0.00602**	(0.00280)	-0.0256***	(0.00518)	-0.00646*	(0.00337)	21,584
UAS 344	-0.00760***	(0.00255)	-0.0239***	(0.00529)	-0.00874***	(0.00314)	21,584
UAS 346	-0.00531*	(0.00303)	-0.0281***	(0.00506)	-0.00427	(0.00399)	21,584
UAS 350	0.00116	(0.00364)	-0.0201***	(0.00562)	-0.00796**	(0.00325)	21,584
UAS 351	-0.00506*	(0.00292)	-0.0197***	(0.00566)	-0.00949***	(0.00306)	21,584
UAS 461	-0.00672**	(0.00267)	-0.0250***	(0.00524)	-0.00575	(0.00356)	21,584
UAS 475	-0.00760***	(0.00255)	-0.0295***	(0.00487)	-0.00938***	(0.00305)	21,584
UAS 502	-0.00301	(0.00333)	-0.0277***	(0.00504)	-0.00752**	(0.00335)	21,584
UAS 559	-0.00387	(0.00324)	-0.0265***	(0.00519)	-0.00732**	(0.00343)	21,584
Working	-0.000745	(0.000779)	-0.00133	(0.00130)	-0.00116	(0.000881)	21,584
Gender	-5.99e-05	(0.000799)	-0.00149	(0.00140)	0.000844	(0.000931)	21,584
White							21,584
Black	-0.00195**	(0.000795)	0.000482	(0.00207)	-0.00130	(0.00118)	21,584
Hispanic/Latino	0.000435	(0.00107)	-9.55e-05	(0.00182)	-0.000574	(0.00129)	21,584
Asian/Other	-0.00182**	(0.000854)	-0.00102	(0.00235)	-0.00297**	(0.00117)	21,584
Married/ Living w/ Partner	-0.000516	(0.000865)	0.00107	(0.00133)	0.000126	(0.000913)	21,584
< High School							21,584
High School Degree	0.00125	(0.000762)	-0.00107	(0.00171)	-0.000406	(0.00109)	21,584
College Degree	0.00238**	(0.000951)	-0.00495***	(0.00176)	-0.000630	(0.00130)	21,584
Age	9.47e-05	(0.000413)	0.000396	(0.000640)	-6.32e-05	(0.000486)	21,584
Medical Condition	0.00174**	(0.000797)	-0.000446	(0.00122)	0.000340	(0.000907)	21,584
Rural Environment							21,584
Mixed Environment	0.00246	(0.00193)	-0.000606	(0.00189)	-0.000463	(0.00123)	21,584
Urban Environment	0.00889*	(0.00455)	-0.00202	(0.00195)	-0.000375	(0.00142)	21,584
Missing Urbanicity	0.00612***	(0.00212)	-0.000627	(0.00211)	-0.000988	(0.00160)	21,584
Pre-K to K							21,584
Middle School to Jr High	-0.000663	(0.000912)	-0.000782	(0.00146)	0.000990	(0.000871)	21,584
High School	-0.00209**	(0.000812)	-0.00255*	(0.00145)	0.00375***	(0.00125)	21,584
Low Income							21,584
Middle Income	0.000732	(0.000890)	-0.00395**	(0.00161)	-0.00298**	(0.00131)	21,584
High Income	0.00109	(0.00113)	-0.00596***	(0.00174)	-0.00409***	(0.00126)	21,584

Number of Kids	0.000105	(0.000380)	-0.000237	(0.000634)	-0.000293	(0.000545)	21,584
Media Trust	-7.63e-05	(0.000484)	0.00116	(0.000871)	0.000254	(0.000597)	21,584
Institutional Trust	-0.000493	(0.000520)	-0.00238**	(0.000932)	-0.00118*	(0.000633)	21,584
Trust in Fox News	0.00107**	(0.000456)	0.00177**	(0.000805)	0.00158***	(0.000511)	21,584
Missing Media Trust	-0.0384***	(0.00587)	-0.109***	(0.0103)	-0.0513***	(0.00668)	21,584
Missing Trump Vote							21,584
Trump Voter	0.00149	(0.000906)	-0.00186	(0.00170)	-0.00134	(0.00133)	21,584

Appendix J: Public Return by Phase

	Private		*Home*		*Other*		*Observations*
UAS 264							4,006
UAS 270	-0.0326**	(0.0150)	0.0509**	(0.0206)	0.0307*	(0.0158)	4,006
UAS 272	-0.0329**	(0.0149)	-0.0146	(0.0168)	0.0123	(0.0143)	4,006
UAS 274	-0.0326**	(0.0152)	0.0174	(0.0195)	-0.00108	(0.0128)	4,006
UAS 276	-0.0364**	(0.0148)	0.00130	(0.0189)	-0.00658	(0.0124)	4,006
UAS 278	-0.0361**	(0.0147)	-0.00449	(0.0186)	-0.00236	(0.0128)	4,006
UAS 280	-0.0293*	(0.0155)	-0.0181	(0.0172)	-0.00212	(0.0132)	4,006
UAS 282	-0.0244	(0.0160)	-0.0214	(0.0167)	-0.0100	(0.0119)	4,006
UAS 340	-0.0318**	(0.0153)	-0.0218	(0.0168)	-0.0183*	(0.0103)	4,006
UAS 342	-0.0363**	(0.0147)	-0.0131	(0.0177)	-0.00107	(0.0133)	4,006
UAS 344	-0.0360**	(0.0149)	-0.0135	(0.0177)	0.00737	(0.0146)	4,006
UAS 346	-0.0304*	(0.0158)	-0.0321**	(0.0159)	-0.00737	(0.0129)	4,006
UAS 350	0.0145	(0.0203)	0.0444*	(0.0234)	0.0328*	(0.0182)	4,006
UAS 351	-0.0257	(0.0167)	0.0608**	(0.0250)	-0.00725	(0.0130)	4,006
UAS 461	-0.0212	(0.0180)	0.0228	(0.0232)	0.00442	(0.0161)	4,006
UAS 475	-0.0378**	(0.0152)	-0.0189	(0.0188)	-0.00723	(0.0144)	4,006
UAS 502	0.0191	(0.0233)	0.0220	(0.0238)	0.0224	(0.0185)	4,006
UAS 559	-0.0250	(0.0175)	-0.0249	(0.0183)	-0.00729	(0.0143)	4,006
Working	-0.00523	(0.00502)	0.00174	(0.00767)	-0.00109	(0.00594)	4,006
Gender	0.000971	(0.00488)	-0.0231***	(0.00815)	-0.0108*	(0.00609)	4,006
White							4,006
Black	-0.0107	(0.00691)	0.0240*	(0.0144)	0.0132	(0.0105)	4,006
Hispanic/Latino	-0.00185	(0.00600)	0.0174	(0.0111)	0.0144	(0.00933)	4,006
Asian/Other	0.00276	(0.00956)	-0.00450	(0.0124)	6.62e-05	(0.00960)	4,006
Married/Living w/ Partner	0.00542	(0.00673)	0.00562	(0.00797)	-0.000112	(0.00660)	4,006
< High School							4,006
High School Degree	-0.00215	(0.00784)	-0.0314***	(0.0113)	-0.00897	(0.00720)	4,006
College Degree	-0.00467	(0.00809)	-0.0542***	(0.0121)	-0.00880	(0.00852)	4,006
Age	-0.00223	(0.00271)	-0.000260	(0.00349)	-0.000953	(0.00268)	4,006
Medical Condition	0.00906*	(0.00480)	-0.00696	(0.00781)	-0.000480	(0.00568)	4,006
Rural Environment							4,006
Mixed Environment	0.0259*	(0.0147)	-0.00811	(0.0117)	-0.0105	(0.00986)	4,006
Urban Environment	0.0347**	(0.0169)	-0.00743	(0.0133)	-0.0237**	(0.0104)	4,006
Missing Urbanicity	0.0388**	(0.0154)	-0.00522	(0.0138)	-0.0180*	(0.0102)	4,006
Pre-k to K							4,006
Middle School to Jr High	0.00546	(0.00578)	0.0347***	(0.00975)	0.00964	(0.00686)	4,006
High School	-0.00222	(0.00553)	0.00859	(0.00882)	0.00717	(0.00703)	4,006
Low Income							4,006
Middle Income	-0.00176	(0.00679)	-0.00780	(0.00999)	-0.0111	(0.00743)	4,006
High Income	0.00641	(0.00795)	-0.0277**	(0.0108)	-0.0147*	(0.00836)	4,006
Number of Kids	-0.000943	(0.00244)	-0.00317	(0.00329)	-0.00180	(0.00280)	4,006
Media Trust	0.00795**	(0.00399)	-0.00459	(0.00592)	-0.00165	(0.00441)	4,006

Institutional Trust	-0.00551	(0.00417)	-0.00634	(0.00610)	-0.00266	(0.00437)	4,006
Trust in Fox News	-0.00119	(0.00299)	0.00289	(0.00503)	0.00524	(0.00337)	4,006
Missing Media Trust							4,006
Missing Trump Vote							4,006
Trump Voter	0.00542	(0.00677)	-0.0386***	(0.0112)	-0.0200**	(0.00882)	4,006

Figure 4: Urbanicity by School Type

*** p<0.01, ** p<0.05, * p<0.1; Marginal Effects as percentage points using Set A